

Geometry and analysis

Student investigations / Remaining atlas entries / September 2026

Choose an investigation whose tools you know. Make things, try ideas and explain what you notice. Use scrap paper or work with a partner. Later pages can be another meeting.

GA-01	Can a box keep a secret?	5
	Fixed-axis spatial tracking; shortest-word proof; no arithmetic needed	
GA-02	Design a fence that defeats a match	8
	Height comparison and continuity; exact fractions are optional	
GA-03	A cover with a tiny budget	11
	Fractions, geometric sums, and reasoning about an infinite list	
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	Half angles and continuous choices; complex notation optional	
GA-05	Can a network hide a high point?	17
	Averages and halves; proof by following a maximum through a graph	
GA-06	What does a real picture miss?	20
	Complex arithmetic and factorization; four-real-dimensional reasoning	
GA-07	Build an angle machine	23
	Polynomial algebra, unit-circle cosine, and cosine addition	
GA-08	Can a rate law choose a future?	26
	Calculus required: derivatives, piecewise joins, and square roots	

Print selected pages, single-sided, on US Letter at actual size. Keep student pages and solutions separate. The original random-ten book remains a separate volume.

Contents continued

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The listed tools are prerequisite gates, not age labels. More detailed gates appear in the facilitator guide.

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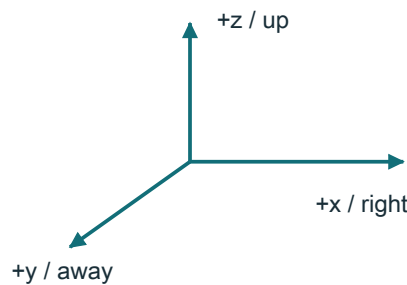
Print selected pages, single-sided, on US Letter at actual size. Keep student pages and solutions separate. The original random-ten book remains a separate volume.

Choose two turns

Can a box keep a secret?

Core: fixed axes and spatial tracking

Reset: the red R face points $+x$ and the blue B face points $+z$. A face points outward, perpendicular to itself. Room axes stay still. X sends $+y$ to $+z$ and $+z$ to $-y$; Y sends $+z$ to $+x$ and $+x$ to $-z$. Both are quarter-turns. A minus card undoes its matching turn. Read a word left to right.



ROOM AXES STAY FIXED

Reset faces: R toward $+x$; B toward $+z$.

X

Y

X-

Y-

1. Choose two cards from X, Y, X-, Y-. Predict where red and blue will point. Carry out your word, reset, then reverse its order. Record one comparison worth showing another pair.

My word / reversed word: red and blue predictions

2. Give your partner an exact reset and two short words that look interchangeable but are not. What single observation disproves equality of the two motions?

Reset / words / final direction that proves a difference

One mark returns

Can a box keep a secret?

Core continuation: inverse moves and a shortest-word argument

Use only X, X⁻, Y and Y⁻. A word may repeat cards. Restore the reset position before every attempt. The challenge concerns the whole orientation, not the box location.

3. Find a nonempty word that returns blue to +z but leaves red pointing somewhere else. Then try to use as few cards as possible. Draw or write your best word.

My word, read left to right

--	--	--	--	--	--

4. Why can no one-card or two-card word win this challenge? Check kinds of words instead of writing all sixteen pairs.

One-card words / same-axis pairs / different-axis pairs

5. A partner says, "Blue ended in the same place, so these motions are the same." Repair the test. How many marked directions are enough for a rigid box?

A test of the whole orientation

What if the turns are flat?

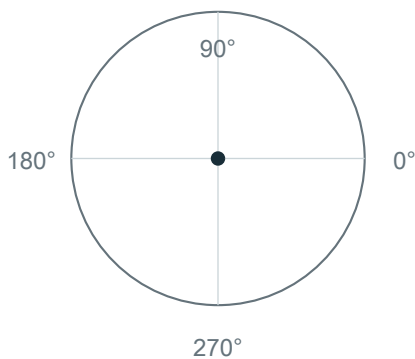
Can a box keep a secret?

Continuation: angles; signed coordinates optional

Use the supplied flat compass for same-center turns and the line with centers $A=0$ and $B=2$ for half-turns. A half-turn about c sends a point x on the line to $2c-x$.

6. Choose two angles about the same center, in either direction. Compare both orders. Explain what happens for any two angles, not just yours.

MY TWO ANGLES



Same-center explanation

7. Start a dot at $A=0$. Compare a half-turn about A followed by a half-turn about $B=2$, with the reverse order. Which word in your same-center claim matters?



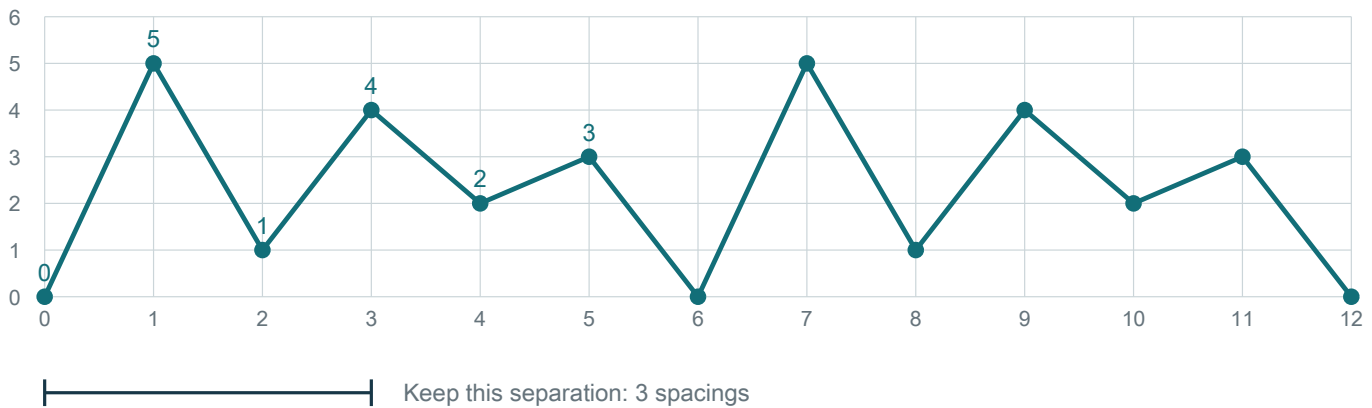
A then B / B then A / the assumption that matters

Try to beat the markers

Design a fence that defeats a match

Core: comparison and opposite positions

The numbered posts are equally spaced. Join neighboring tops by straight ramps, and repeat the profile after six spacings. Red and blue are always exactly three spacings apart. Slide both in the same direction; corners are allowed.



1. On the supplied fence, choose a starting position that is not a match. Move the markers and mark a place where their heights seem equal. Explain how you kept them opposite.

2. Choose six new heights on the blank grid. Make every pair of opposite posts unequal. Join all ramps, including the seam. Try to design a fence whose moving opposite markers never match. Keep one attempted design and mark what happened.



Repeat the first height at post 6.

3 spacings on this grid

Explain the exchange

Design a fence that defeats a match

Core proof: continuity and a half-turn

Imagine that red begins higher. The markers move half a turn, without lifting or jumping, until each reaches the other marker's starting place.

3. What must be true about which marker is higher at the finish? Explain why a match must occur somewhere on the way. Include the case in which the start is already a match.

START: red higher

AFTER HALF A TURN

Why the change must pass through a match

4. Another group checked all six posts and found no equal opposite heights. Does that defeat your argument? Use the printed fence to settle the disagreement.

Posts / positions between posts

Test the rules

Design a fence that defeats a match

Continuation: exact locations use fractions

For exact work let x measure post spacings, with $x=0$ at the height-0 post. Opposite locations differ by 3; x and $x+6$ name the same position. For the rule tests, give a precise counterexample or explanation.

5. Keep opposition but allow jumps. On one half of the circle set height 0 and on the other half set height 1: $h(t)=0$ for $0 \leq t < 1/2$ and $h(t)=1$ for $1/2 \leq t < 1$, repeated every turn. Can opposite heights match? Where did the proof break?

At the middle and at the seam

6. Optional exact challenge: find every matching pair on the printed straight-ramp fence. Work over $0 \leq x < 3$ so you do not count the same opposite pair twice.

Three ramp intervals: choose u and compare the two heights

7. Replace the circle by a straight trail: height $h(t)=t$ for $0 \leq t \leq 1$. Keep walkers $1/2$ unit apart, with the first at $0 \leq t \leq 1/2$. Must they ever match? Explain which part of the circle argument is lost.

What the trail example shows

Name a point; buy a cover

A cover with a tiny budget

Core entry: fractions and lengths

An interval has a center and a positive width. It covers its center even when the interval is open (endpoints excluded). You pay the sum of assigned widths; overlap does not refund your budget. The opponent starts with 0, 1, $1/2$, then chooses two more points in $[0, 1]$.

1. Choose widths and cover these five points for a total cost less than $1/100$. Describe each interval by its center and width. Challenge a partner with your fourth and fifth points.

Choose two additional points.



Centers and positive widths; cost of all five

2. The opponent now promises to keep naming points forever. Before hearing them, invent a rule giving a positive width to point number n . It must keep the total cost below $1/100$, no matter which points are named. Test several terms without pretending the test proves the whole rule.

My rule for point number n / why the future still has a budget

Defend the promise

A cover with a tiny budget

Core: a rule for every numbered point

Keep your rule visible. A list may repeat points and may be ordered by an opponent. Your explanation must work for a point very far down the list and for every positive budget, not merely for the first picture.

3. Prove both parts of your promise: every listed point is covered, and the total cost stays below $1/100$. Write a finite partial-sum formula or a geometric argument that also controls the whole tail.

Coverage of p_n / partial sum and tail

4. An opponent changes the budget to a positive number B as small as they like. Adapt your rule. Does “a cover for every $B > 0$ ” mean you have one cover whose intervals all have width zero?

My rule for a new positive budget B

5. Suppose instead that you pay the same positive width w for each named point forever. Can that assigned-cost rule fit under a fixed finite budget? Would overlapping intervals change this answer?

Assigned cost and union length are different quantities

The points become dense

A cover with a tiny budget

Advanced continuation: fractions, lists and a supplied theorem

List fractions row by row: for $q=1,2,3,\dots$ list $0/q, 1/q, \dots, q/q$. Repetitions are allowed. Supplied theorem: any countable collection of open intervals covering every real point of $[0,1]$ has summed lengths at least 1. We use this theorem; a finite drawing is not its proof.

6. Choose three fractions in $[0,1]$, including one with a large denominator. Find the row and position at which each appears. Explain why every rational number in $[0,1]$ appears at a finite place. Can the tiny-budget rule cover this entire list?

$q=1$	0/1	1/1				
$q=2$	0/2	1/2	2/2			
$q=3$	0/3	1/3	2/3	3/3		
$q=4$	0/4	1/4	2/4	3/4	4/4	
$q=5$	0/5	1/5	2/5	3/5	4/5	5/5

My three fractions and where they appear

7. There are rationals between any two different real points. Does covering them all with total cost below $1/100$ force your intervals to cover every real point too? Use the supplied theorem. What can you say about a point that is left out?

Use the supplied theorem; distinguish existence from a formula

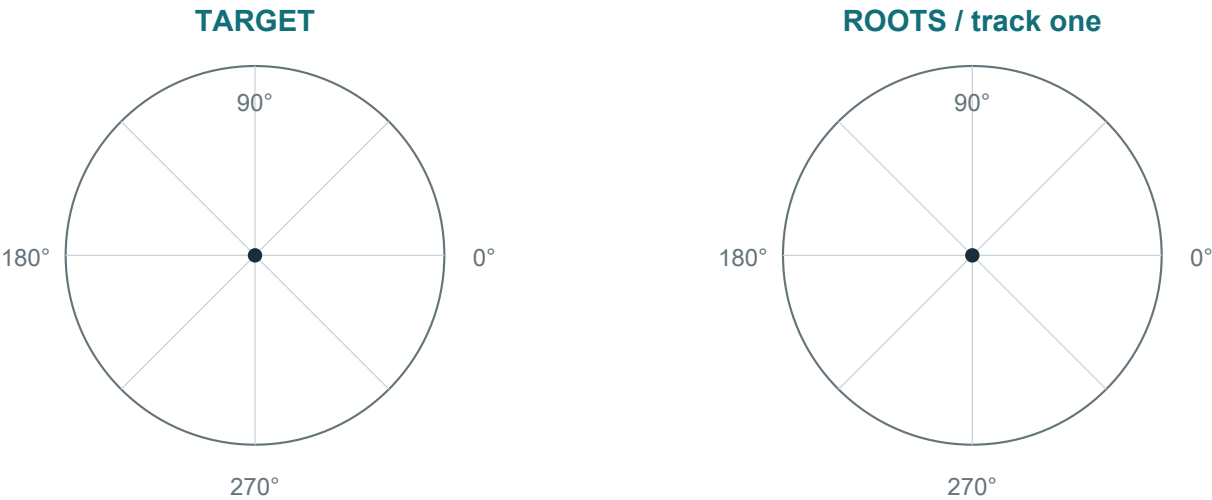
Choose a route

A root takes a journey

Core: angle doubling and halving

A root arrow is legal when doubling its angle gives the target angle, modulo one full turn. There are two legal unit root arrows for each unit target arrow. Start the target and chosen root at 0°. You may turn either way, but the chosen root must move continuously and remain legal at every instant.

1. Choose a target position that is not 0° or 180°. Find both legal roots and explain why there are exactly two. Test your pair by doubling the angles.



2. Invent a target route starting at running angle 0° and finishing at 360°. You may stop or reverse along the way. Record a few signed moves, then follow one root without jumping. Predict and check its final direction.

My signed target moves				
My signed root moves				

My endpoint prediction and check

Can another choice escape?

A root takes a journey

Core proof: continuity

Compare two possible no-jump root choices along the same target route. At every moment they are either the same root or opposite roots. A drawing of selected moments suggests an answer; explain what happens between them.

3. Could a clever root follow your choice for a while, switch to the other root without a jump, and return to the start after one target turn? Explain why or construct such a motion.

Could two separated legal alternatives exchange continuously?

4. Suppose someone claims to assign one root to every target position on the circle, with the assigned root varying continuously as the target moves. Use one circuit to test the claim.

The same target at start and finish

Count the net turns

A root takes a journey

Continuation: integer turns; optional mth roots

For these tasks routes are supplied by a continuous running angle θ , with $\theta(\text{start})=0$ and $\theta(\text{finish})=360^\circ k$ for an integer k . This is our definition of k net turns for these routes; no theorem about arbitrary loops is needed.

5. Choose one route with reversals that restores the root and one that does not. Give a rule using k that decides all these routes, including clockwise circuits.

My signed target moves				
My signed root moves				

My signed target moves				
My signed root moves				

Return rule, including negative k

6. Advanced: change the machine so that it triples a root angle. Start with the root at 0° . Find all legal roots for target 120° , and decide how many counterclockwise target circuits first restore your tracked root.

Legal roots / first return / explanation

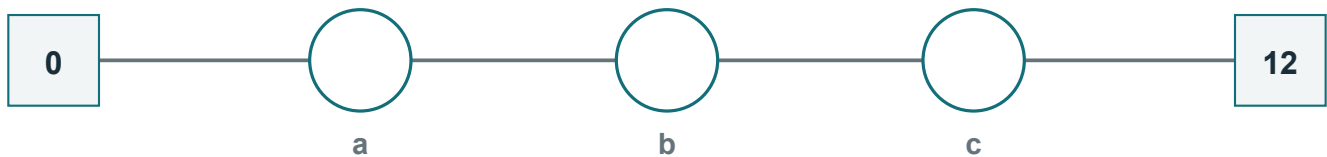
Change one connection

Can a network hide a high point?

Core: averages and halves

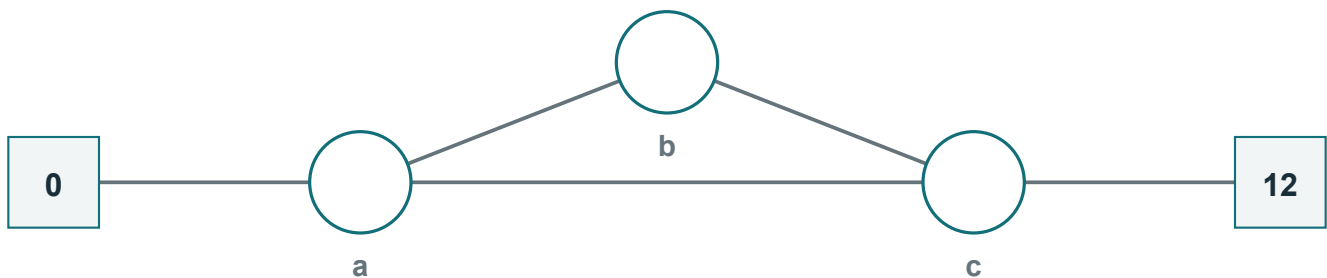
At every circle, its number equals the average of all adjacent numbers. Squares are fixed and do not obey this rule. All edges count once with equal weight, regardless of drawing length. The original network is 0-a-b-c-12.

1. Choose numbers for a,b,c that make every circle obey the rule at the same time. Check your whole assignment. Can equal steps along the path help?



Check all three averages

2. Now connect a directly to c, keeping every old edge. Before solving, predict which of a,b,c will rise, fall or stay. Find a legal new assignment and explain your prediction.



Prediction / new values / simultaneous check

Build a hiding place

Can a network hide a high point?

Core proof: the greatest-value argument

Draw a finite network of at most six nodes. Choose at least one fixed square in every connected component; use fixed values between 0 and 12. Circles must obey the same averaging rule. Your challenge is to make some circle strictly above 12.

3. Try to design a network and legal values that hide a circle above 12. Keep a record of one attempt. If an average fails, mark the failed circle instead of silently changing the rule.

My attempted graph: square fixed nodes, circle averaging nodes

4. Assume a legal network has a largest value M greater than every fixed value. At a circle with value M , what must all its neighbors equal? Follow a path to a square and finish the argument. Also explain the lower bound.

Follow a largest value along a path to a fixed square

5. Break exactly the boundary condition: add two circles joined to each other, disconnected from every square. Can this pair carry 100 while the old network stays legal? What claim fails?

A component without a fixed square

Could two answers both work?

Can a network hide a high point?

Proof continuation: subtraction of assignments

Use any finite graph with a nonempty fixed boundary in each component. A “solution” means all circles obey the rule simultaneously. You may reuse the maximum and minimum result from the previous page.

6. Suppose two different assignments solve exactly the same network and fixed values. At each node subtract the second number from the first. What rule do these differences obey? Decide whether both assignments can exist.

What subtraction does to the boundary and averaging rules

7. On the shortcut network, raise only the right square from 12 to 16. Before solving, bound how far any circle can move. Then find the three new values and test your bound.

Bound first / then solve / compare the changes

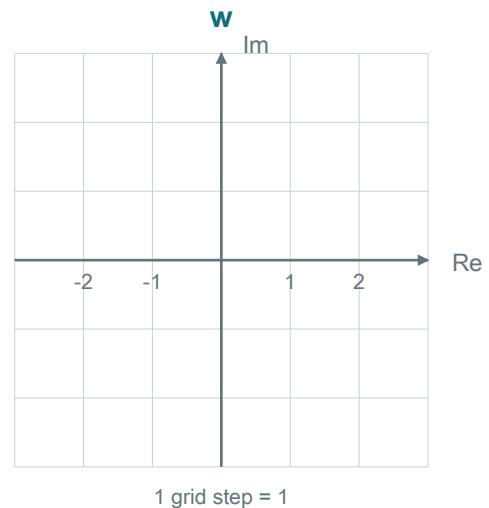
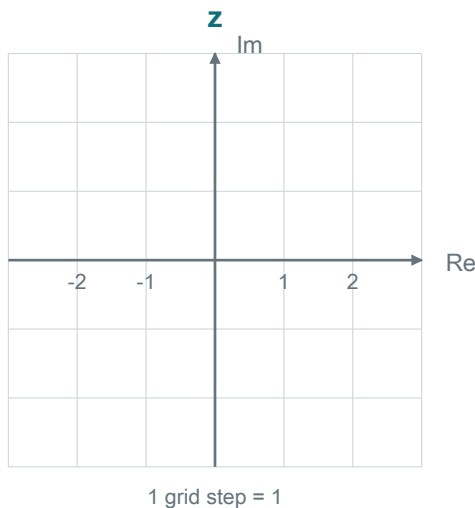
Choose the coordinates

What does a real picture miss?

Core: complex algebra

Solve $z^2 + w^2 = 0$ over the complex numbers. A point in this problem is a pair (z, w) , not a single point on one ordinary graph. The two Argand grids record its two coordinates.

1. Choose $z=1+i$ or $z=2-i$, then find every possible w . Plot each linked pair on the two grids. Challenge a partner with a third z of your own.



2. Find a formula describing every solution pair. Prove that no others are missed, and determine exactly where your two formulas give the same pair.

All solutions / no missing cases / where branches meet

Restrict the view

What does a real picture miss?

Core continuation: real slices; path question uses continuity

The two branches each have a freely chosen complex parameter. A restriction to real values of some coordinates is a slice. It is not a picture of every complex solution.

3. Find all solutions when both z and w must be real. Then allow z to stay real but w to be complex. Describe both slices and explain why the first picture could be misleading.

Both coordinates real

z real, w complex

4. Begin at $(1, i)$ and try to move continuously through solutions to $(1, -i)$, without ever passing through $(0, 0)$. Can a route exist? If you allow the origin, give an exact route.

Without the origin / an exact route if the origin is allowed

Change the equation

What does a real picture miss?

Advanced algebra continuation: an invertible complex change of coordinates

Set $u=w-iz$ and $v=w+iz$. Both new coordinates are still complex. Investigate $z^2+w^2=1$ with this new notation; the equation no longer asks for a product to be zero.

5. Compute uv , then recover z and w from u and v . Use this to give every solution of $z^2+w^2=1$ with one freely chosen parameter. State any forbidden parameter value.

Product / inverse coordinate formulas / parameter restriction

6. Choose $u=2$ or $u=i$ and construct the corresponding (z,w) . Can you give a continuous route through constant-one solutions from $u=1$ to $u=-1$ without meeting a branch junction?

Construct a pair / verify the equation / describe a continuous route

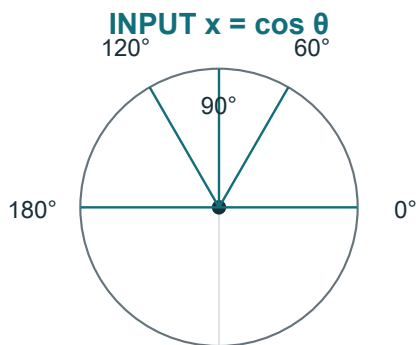
Choose useful tests

Build an angle machine

Core: unit-circle cosine and algebra

The input x is the horizontal coordinate of an angle arrow: $x = \cos \theta$. The desired output is $\cos(3\theta)$. Start by searching among odd cubics $P(x) = ax^3 + bx$. The printed rays include $0^\circ, 60^\circ, 90^\circ, 120^\circ, 180^\circ$; select tests deliberately.

1. Choose two of the printed angles and use them to determine a and b , if you can. If your tests do not determine them, explain why and choose a more useful pair.



My chosen tests and equations

Candidate: $P(x) =$

2. Choose another angle to challenge your candidate. Also compare θ and $-\theta$: do equal inputs ask the machine for conflicting outputs? State what these checks do and do not prove.

A new test / equal-input check / limits of the evidence

From evidence to an identity

Build an angle machine

Core proof: cosine addition and polynomial recurrence

You may use $\cos(A+B)+\cos(A-B)=2\cos A \cos B$. Define $T_0(x)=1$, $T_1(x)=x$, and $T_{n+1}(x)=2xT_n(x)-T_{n-1}(x)$ for $n \geq 1$.

3. Use the recurrence to build T_2 and T_3 . Then explain why $T_n(\cos \theta)=\cos(n\theta)$ for every integer $n \geq 0$ and every real θ .

Build the first polynomials / justify the induction step

4. A rival adds $x(x^2-1)(x^2-1/4)$ to your cubic. Show why the new polynomial passes all five original angle tests but still fails somewhere. What distinguishes a fitted guess from your proof?

Why five tests pass / a failing input / the role of proof

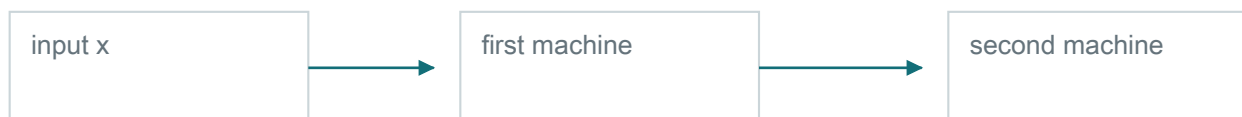
Feed one machine into another

Build an angle machine

Advanced continuation: composition and polynomial identities

A doubling machine has polynomial T_2 ; a tripling machine has T_3 . Feeding one machine's output into the next means substituting one polynomial for x in the other.

5. Predict what $T_2(T_3(x))$ and $T_3(T_2(x))$ do when $x = \cos \theta$. Choose one order to expand, and compare with a partner who expands the other.



My expansion and the other order

6. State and prove a rule for $T_m(T_n(x))$. First work on $-1 \leq x \leq 1$. Supplied algebra fact: a nonzero polynomial of degree d has at most d distinct real roots. Can your rule extend outside the circle-input interval?

First on the cosine interval / then as a polynomial identity

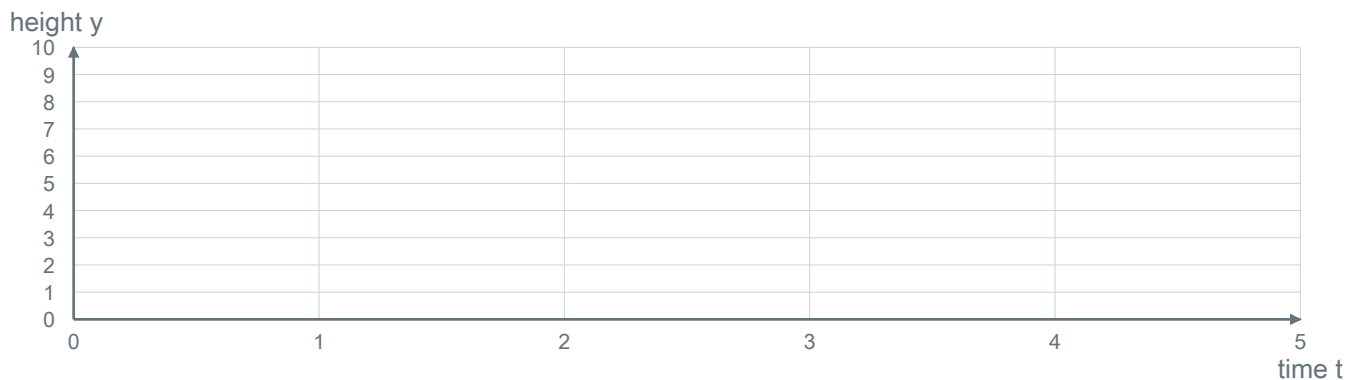
Invent a motion

Can a rate law choose a future?

Calculus core: derivatives of powers

A solution must be defined for all $t \geq 0$, be nonnegative, have $y(0)=0$, and satisfy $y'(t)=2\sqrt{y(t)}$ at every time. Use the right derivative at $t=0$. The equation specifies an instantaneous rate, not an update step.

1. Search among $y(t)=a t^2$ and $y(t)=a t^4$ with $a \geq 0$. Which choices of a obey the rate law for every time? Use them to find different futures from the same start.



Exact checks for every t , including $t = 0$

2. Both of your valid futures have slope 0 at the start. Explain why that fact does not force the graph to remain flat. What would count as a complete check of another proposed future?

One instantaneous slope / a complete solution check

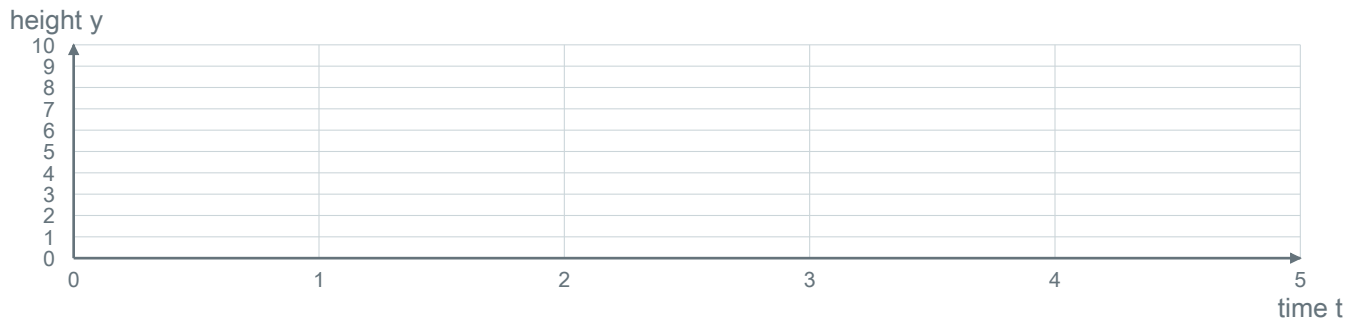
Choose when to leave

Can a rate law choose a future?

Calculus core: piecewise functions and the join

Choose a waiting time $c > 0$, for example $3/2$ or 2 , and require $y(t) = 0$ until $t = c$. Your job is to invent the rest of the function and defend the exact moment at which it starts moving.

3. Find a formula for a motion that waits until your c , then becomes positive. Verify the rate law on both open pieces $t < c$ and $t > c$.



My c and my piecewise formula

4. At $t = c$, calculate the difference quotient $[y(c+h) - y(c)]/h$ separately for $h < 0$ and $h > 0$. Is the joined motion differentiable, and does it obey the rate law at the join?

$h < 0$: difference quotient and limit

$h > 0$: difference quotient and limit

5. A partner tries the single formula $y = (t - c)^2$ for all $t \geq 0$, with $c > 0$, avoiding a piecewise definition. Identify every obstruction to calling this a solution of our initial-value problem.

Initial condition / signs before c / equation after c

Are there stranger futures?

Can a rate law choose a future?

Advanced calculus continuation: chain rule and monotonicity

Consider any nonnegative differentiable solution on all $t \geq 0$. You may use: a function with nonnegative derivative is nondecreasing, and a function with derivative 1 on an interval equals t plus a constant.

6. Could a solution leave zero, return to zero, and leave again? Explain the possible shape of its set of zero times.

Sign of the derivative / possible zero times

7. On any interval where $y > 0$, set $v(t) = \sqrt{y(t)}$. Find v' . Use continuity at the last zero c to classify every nonnegative solution of the original problem.

Positive interval / chain rule / continuity at c / all cases

What changes the answer?

Can a rate law choose a future?

Advanced calculus: product rule and a stated uniqueness hypothesis

For a scalar rate function f , local Lipschitz at 0 would require constants $L, \delta > 0$ with $|f(a) - f(b)| \leq L|a - b|$ for all permitted a, b within δ of 0. A standard uniqueness theorem uses local Lipschitz continuity (plus continuity in time). We inspect the hypothesis here, not prove that theorem.

8. Change the rate law to $y' = y$ with $y(0) = 0$. Differentiate $e^{-t}y(t)$ and determine every solution. Does a delayed departure remain possible?

Differentiate the product and use the initial value

9. For $f(y) = 2\sqrt{y}$ on $y \geq 0$, compare $f(h) - f(0)$ with h as $h > 0$ tends to 0. Then test the claim “every non-Lipschitz rate gives several nonnegative forward futures” using $y' = -2\sqrt{y}$, $y(0) = 0$.

Ratio near zero / negative-rate counterexample / logical conclusion

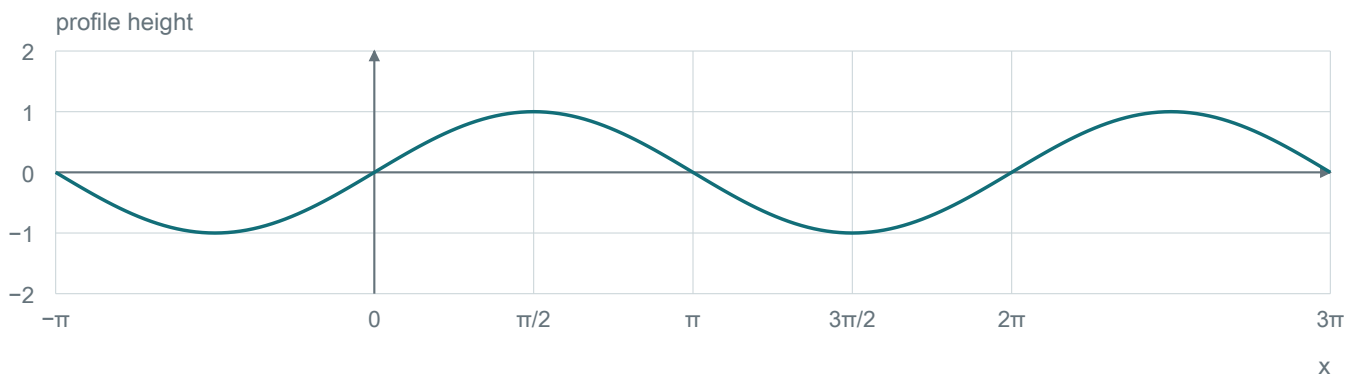
Slide it or fade it

Choose a shape, then choose its future

Core: partial derivatives and trigonometric functions

For all real x and $t \geq 0$, try $U_c(x,t) = \sin(x-ct)$ and $V_a(x,t) = e^{(-at)} \sin x$. Choose c from $-2, -1, 0, 1, 2$ and a from $0, 1, 4$. The two candidate laws are $W_{tt} = W_{xx}$ (wave) and $W_t = W_{xx}$ (heat). A subscript is a partial derivative; hold the other variable fixed. Angles use radians.

1. Choose c and a . Draw the profiles at $t=0$ and $t=\pi/2$. Predict what happens to the peak position and height. Choose a space-time point where you would test a disputed prediction.



Teal curve: initial $\sin x$. My choices: $c = \underline{\hspace{1cm}}$ $a = \underline{\hspace{1cm}}$. Add later profiles.

2. Classify all the allowed c and a : which U_c and V_a obey each law everywhere? Prove each success and exhibit a failed equality for each excluded type.

Derivatives / legal parameters / counterexamples

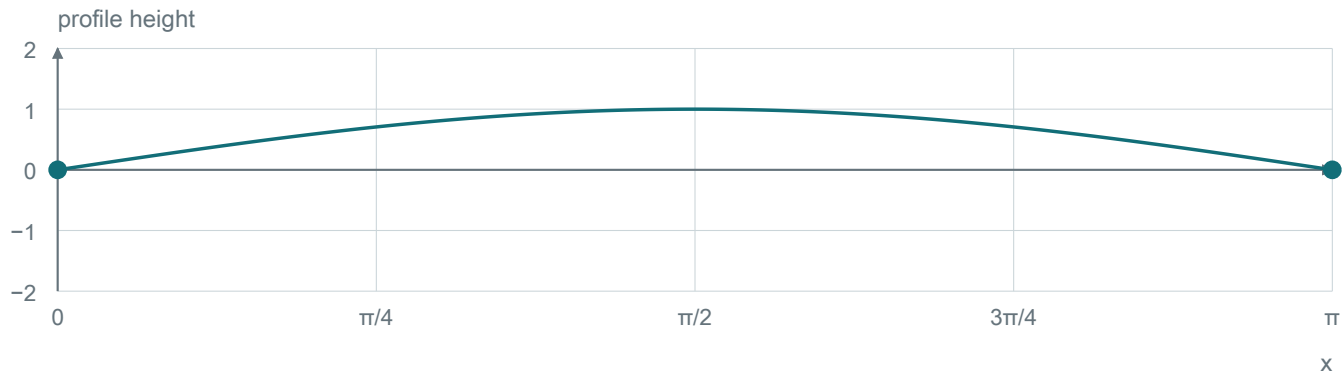
One initial shape, several futures

Choose a shape, then choose its future

Core continuation: initial and boundary data

Now use only $0 \leq x \leq \pi$ and require $W(0,t)=W(\pi,t)=0$ for every $t \geq 0$. The initial shape is $\sin x$. Try $W_b(x,t) = \sin x(\cos t + b \sin t)$, with a real number b of your choice. The heat candidate is $H(x,t) = e^{-t} \sin x$.

3. Choose two different values of b . Verify that both W_b satisfy the wave law, the endpoint rule and the same initial shape. Show that their later futures differ. Explain why simply sliding $\sin(x-t)$ would not meet this endpoint rule.



Initial shape shown. My b values: _____ and _____. Add and label later profiles.

Wave law / boundary check / initial data

4. Choose b so the wave W_b and heat H have both the same initial shape and the same initial velocity. Find an exact later observation that separates them. What additional choice remains essential when specifying a future?

Match the initial velocity / choose a later observation

Design with more than one mode

Choose a shape, then choose its future

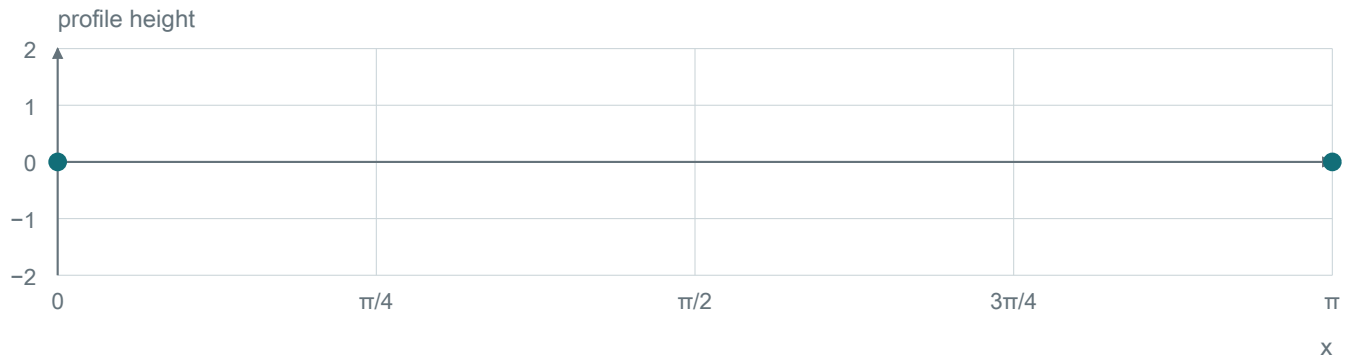
Calculus continuation: separated modes and linearity

A mode has spatial shape $\sin(kx)$, where k is a positive integer. Test heat candidates $e^{(-\alpha t)}\sin(kx)$ with $\alpha \geq 0$ and wave candidates $\cos(\omega t)\sin(kx)$ with $\omega > 0$, keeping the fixed endpoints.

5. Choose k . Determine every α and ω for which the corresponding candidate obeys its law. Explain how changing the number of spatial oscillations changes decay and oscillation in time.

My k / time factors / derivative comparison

6. Choose a nonzero real d and initial shape $\sin x + d \sin(2x)$. Build a heat evolution and, separately, a wave evolution with zero initial velocity from the modes. Can the heat evolution at any positive time be just one scalar multiple of the whole initial shape? Prove your answer.



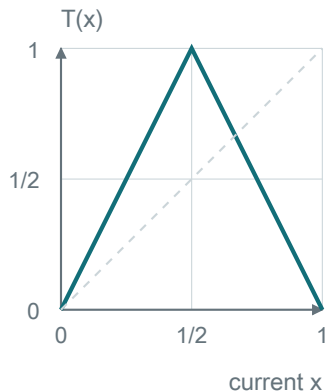
My nonzero d / two evolutions / scalar-multiple test

Write a program before guessing a start

Program the folded interval

Core: exact fractions and a folded rule

$T(x)=2x$ when $0 \leq x \leq 1/2$, and $T(x)=2-2x$ when $1/2 \leq x \leq 1$. The formulas agree at $1/2$. Record L for a current state in $[0, 1/2)$, and R for a current state in $[1/2, 1]$. A program lists current-state labels, before each update.



My chosen four-letter program

Current half, before each update

1. Choose a four-letter word using L and R. Try exact starting fractions to make those four labels repeat. Keep a successful orbit or explain precisely where a trial fails.

Exact states and current-half labels

2. Test starts $1/2$, 1 and 0 . Can a genuinely periodic orbit ever pass through $1/2$ or 1 ? Explain what this means for the L/R boundary convention.

Boundary trajectories / periodic-orbit explanation

Undo the machine

Program the folded interval

Core continuation: inverse branches and composition

For a target $y \in [0, 1]$, its two possible previous values are $L(y) = y/2$ and $R(y) = 1 - y/2$. Here L and R name inverse cards as well as the corresponding half. For word $s_0 s_1 s_2 s_3$, a return candidate solves $x = B_{s_0}(B_{s_1}(B_{s_2}(B_{s_3}(x))))$: undo the latest step first.

L $y/2$

R $1 - y/2$

3. Use inverse cards to solve your own four-letter program. Then independently check every forward value, half label and the return. If your first word was too easy, choose another.

My inverse composition / fixed point / forward verification

4. Compare LRRL, LRLR and LLLL. Find their exact starting points and least positive return times. Why is “the word has four letters” not enough to prove least period four?

Programs / exact states / first returns

How many different cycles?

Program the folded interval

Continuation: classification and cyclic starting points

Count geometric cycles, not ways of choosing their starting point. A cycle of least period four contains four distinct states. You may organize words however you wish; no list of answers is supplied.

5. Find every cycle of least period four. Give exact fractions for its states and a way to check that you have not counted the same cycle twice.

Open cycle inventory / how I avoid repeats

6. A partner says that all sixteen four-letter words must produce sixteen four-cycles. Refute the claim, and give a corrected count that separates fixed points, two-cycles and four-cycles.

Correct the count and explain what is being counted

Why backward programs always work

Program the folded interval

Proof continuation: affine fixed points and word periods

For any nonempty word w of length n , let B_w be its inverse-card composition in the same time-order convention. A word has least word period d if it is repeats of a length- d block and no shorter block, where d divides n .

7. Prove that B_w has exactly one fixed point in $[0, 1]$, that this point really follows the repeated word, and that different length- n words give different points. Include the midpoint and endpoint cases.

Existence in $[0, 1]$ / forward labels / distinct starting points

8. Prove that the least orbit period equals the least word period. Use this to justify the completeness of your four-cycle inventory.

Least period argument / completeness of the inventory

Two payment machines

Can tiny payments break a budget?

Core: fractions and chosen targets

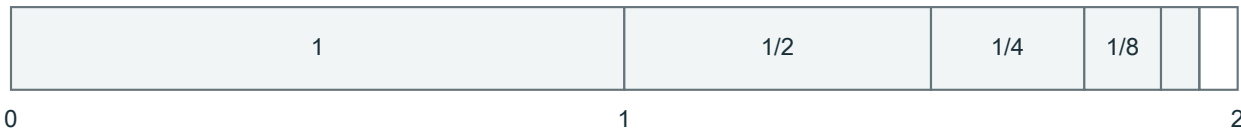
Machine G pays $1, 1/2, 1/4, 1/8, \dots$. Machine H pays $1, 1/2, 1/3, 1/4, \dots$. The total after n payments is G_n or H_n , with $n \geq 1$. A budget is broken only when a total is strictly greater than it.

1. Choose two target budgets, one below 2 and one above 2. Try to break each using each machine. Record what your trials prove and what they leave uncertain.

My target below 2

My target above 2

2. Find a formula for G_n and its exact unpaid distance from 2. For a tolerance ϵ of your choice with $0 < \epsilon < 1$, give a rule selecting n with $2 - G_n < \epsilon$. Test $\epsilon = 1/100$.



Finite formula / gap / my ϵ / strict accuracy test

Build a certificate from groups

Can tiny payments break a budget?

Core continuation: lower bounds, not exact totals

You may group consecutive H payments any way you like. Replacing every payment in a group by a smaller amount gives a lower bound for that group.

3. Use the cards through $1/16$ to prove $H_{16} > 3$ without finding their exact sum. Choose your own grouping and explain why your inequality is strict.

1	$1/2$	$1/3$	$1/4$
$1/5$	$1/6$	$1/7$	$1/8$
$1/9$	$1/10$	$1/11$	$1/12$
$1/13$	$1/14$	$1/15$	$1/16$

My grouping and strict inequality

4. Let a partner choose any finite positive budget B . Give a finite stopping index guaranteed to break it, and prove your rule works for every B . Give one explicit guarantee for $B=10$; do not claim it is the first crossing.

Partner's B / stopping rule / why it always works

Can skipping payments save the budget?

Can tiny payments break a budget?

Continuation: tails and changed rules

A proposed repair ignores the first N payments, with N a positive integer, but keeps every later payment. A different repair keeps only selected indices. These are different operations.

5. After any chosen N , show that the remaining H payments can still break any extra budget. Contrast this with the total of all G payments after its first N payments.

Start after any N / groups / extra budget / comparison with G

6. Choose infinitely many H payment indices so that their total does stay within a finite budget. Prove your choice works. Explain why this does not contradict the previous tail argument or the fact that both machines have payments tending to zero.

An infinite selection / its finite budget / why this changes the rule

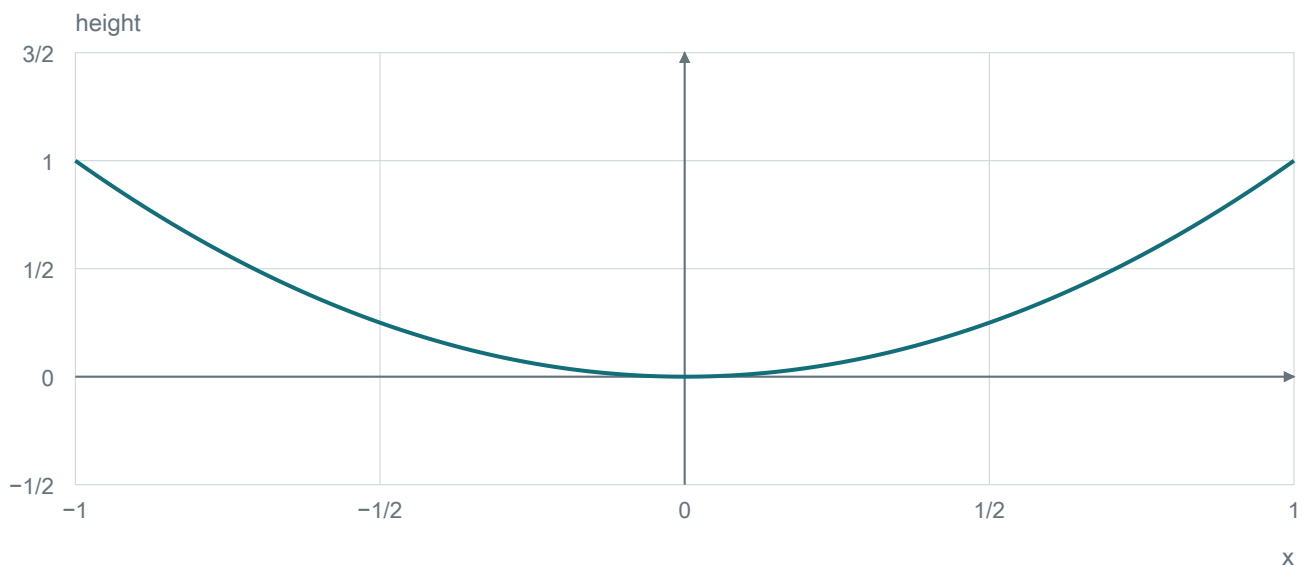
A line and a challenger

How narrow can the error band be?

Core: graphs and vertical error

The target is $f(x)=x^2$ for $-1 \leq x \leq 1$. You may choose any line $p(x)=ax+b$. An error budget $E \geq 0$ is valid if $|x^2 - p(x)| \leq E$ for every x in the interval. The band is between $p-E$ and $p+E$, so its total vertical thickness is $2E$.

1. Choose a line and an error budget. Let a partner challenge them at any x in $[-1,1]$. Revise your proposal and try to make the budget smaller. Mark the actual vertical gaps you compared.



2. A line passes exactly through the two endpoint values of the parabola. Find that line and its worst error on the whole interval. Can endpoint agreement alone certify a small uniform error?

Endpoint line / an interior challenge / whole-interval error

A certificate against every line

How narrow can the error band be?

Core proof: combine inequalities

For a general candidate $p=ax+b$, keep E as an unknown valid error bound. You may test particular x -values to force a lower bound, but must also verify your winning line everywhere.

3. Use $x=-1,0,1$ to prove that no line can have $E<1/2$. Find a line attaining the bound, and verify its error on all of $[-1,1]$.

x	signed residual $x^2-(ax+b)$	consequence of budget E
-1		
0		
1		

Lower bound / attaining line / verification for every x

4. Prove that there is only one line attaining this best budget. Explain why verifying a candidate and proving it is the unique best candidate are separate tasks.

Equality conditions and uniqueness

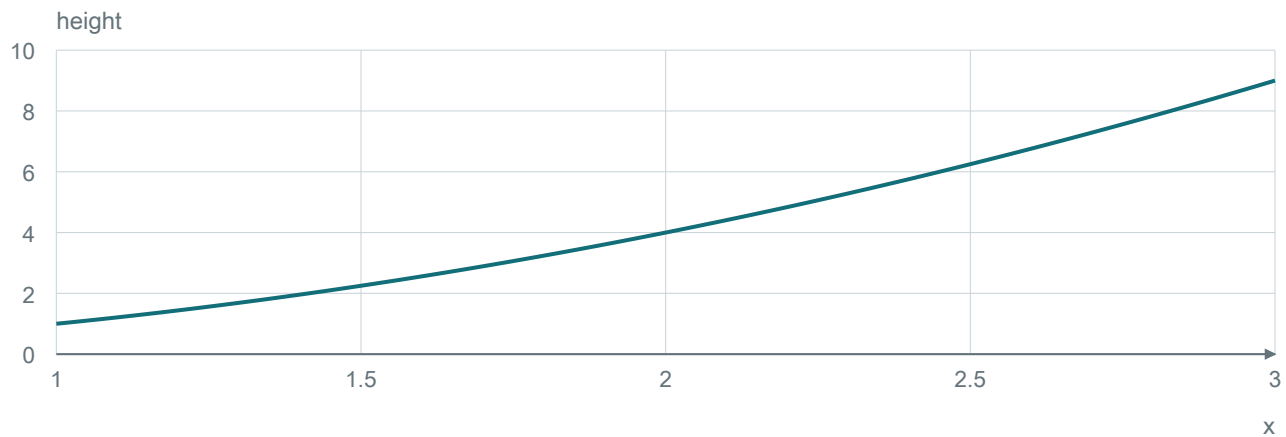
Move and stretch the playing field

How narrow can the error band be?

Algebra continuation: transfer a certificate

Choose a center h and radius $r > 0$. Now approximate x^2 on $[h-r, h+r]$. You may use $z = (x-h)/r$, so $-1 \leq z \leq 1$. Error still means vertical error.

5. Find the best affine line and exact error for your chosen interval. Prove optimality and uniqueness by transferring the earlier argument. Test the interval $[1, 3]$.



The test graph is x^2 on $[1, 3]$. My general interval: $h = \underline{\hspace{1cm}}$, $r = \underline{\hspace{1cm}} > 0$.

6. Compare the optimal line with the line through both endpoint values on your chosen interval. Find the latter line and its exact worst error. Explain how the best line improves it.

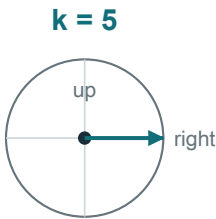
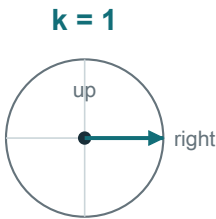
5-6: general certificate / endpoint secant / whole residual range

Make a shared record

Choose a photograph that reveals the motion

Core: unit-circle turns and times

A unit arrow starts pointing right at time 0 and turns counterclockwise. Frequency k means k complete turns each second. Its tip is $(\cos(2\pi kt), \sin(2\pi kt))$; a height-only photograph records $s_k(t) = \sin(2\pi kt)$. Compare $k=1$ and $k=5$.



1. Turn or draw both arrows at $t=0, 1/4, 1/2, 3/4$. Record their heights and full tip positions. Choose another scheduled time $n/4$ and test whether it separates the motions.

t	tip: k=1	tip: k=5	height: k=1	height: k=5
0				
1/4				
1/2				
3/4				

2. Prove agreement at every scheduled time $t=n/4$, not only the ones you drew. Could the entire infinite regular record decide between these two frequencies?

Another scheduled time / proof for every integer n

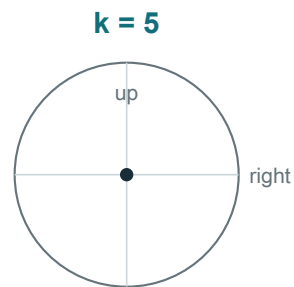
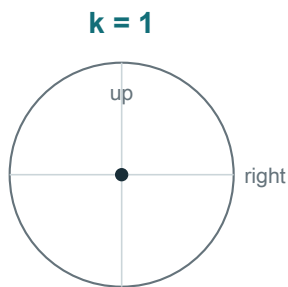
Design one more observation

Choose a photograph that reveals the motion

Core continuation: distinguish a stated class

You may choose one extra observation time between 0 and $1/4$. First distinguish only frequencies 1 and 5. Then broaden the candidate class to all nonnegative integer frequencies with the same starting arrow.

3. Choose an extra time that distinguishes the two heights and justify your choice exactly. Give one tempting extra time that would fail, and explain why.



My extra time / two heights / an extra time that fails

4. Find all nonnegative integer k whose heights agree with frequency 1 at every $t=n/4$. Now also require agreement at $t=1/8$. Does that additional photograph uniquely identify frequency 1 in this enlarged class?

All integer frequencies / necessity / sufficiency / extra photograph

What did the camera forget?

Choose a photograph that reveals the motion

Algebra continuation: sensor choice and finite data

For the next task change the schedule to $t=n/2$, still with nonnegative integer frequencies and arrows initially pointing right. For the final task allow arbitrary continuous real-valued signals, not just rotating-arrow heights.

5. At the new schedule, classify every frequency that looks like $k=1$ when only height is recorded. Then classify them when the full arrow is recorded. Prove whether the richer record removes all ambiguity.

Height record / full-arrow record / which frequencies survive?

6. An observer makes any finite list of measurements at distinct times $t_1, \dots, t_m$. Starting with a continuous signal f , construct a different continuous signal agreeing at every measured time. What claim about unrestricted reconstruction does your construction rule out?

A second continuous signal / matching measurements / why it is different

Mix cards into a picture

Decode a picture from four sign patterns

Core: signed addition and four-square arrays

Read entries in order 00,01,10,11: top row first, then bottom. The cards are $C=(1,1,1,1)$, $R=(1,1,-1,-1)$, $S=(1,-1,1,-1)$, $D=(1,-1,-1,1)$. A weight multiplies every entry of its card. Add contributions square by square; any real weights are allowed.

C	R	S	D																
<table><tr><td>+</td><td>+</td></tr><tr><td>+</td><td>+</td></tr></table>	+	+	+	+	<table><tr><td>+</td><td>+</td></tr><tr><td>-</td><td>-</td></tr></table>	+	+	-	-	<table><tr><td>+</td><td>-</td></tr><tr><td>+</td><td>-</td></tr></table>	+	-	+	-	<table><tr><td>+</td><td>-</td></tr><tr><td>-</td><td>+</td></tr></table>	+	-	-	+
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+	-																		
+	-																		
+	-																		
-	+																		

1. Choose weights for the cards and make a picture. Give your partner only the resulting four entries and ask for the weights. You may use a large constant-card weight to make every final entry nonnegative, even if some other weights are negative.

00	01
10	11

My weights / partner's reconstruction

2. Decode the supplied picture (5,0,1,2). Rebuild all four entries from your weights to verify the result. Would restricting every weight to be a nonnegative whole number change the problem?

Supplied picture

5	0
1	2

Weights / verify every square

Invent a detector for one card

Decode a picture from four sign patterns

Core proof: weighted sums and cancellation

A detector applies a sign to each picture entry, then adds the four signed values. For a sign card P , write $\langle v, P \rangle$ for that weighted sum. No normalization is built into this notation.

3. Choose one card and build a detector that ignores every other card. Explain why it works. Find the detector output on its own card and use it to justify your decoding scale.

Detector signs

00	01
10	11

What my detector does to each card / scale

4. Prove that every real four-entry picture is reconstructed by these four detector readings divided by four, and that no picture has two different weight lists.

Arbitrary entries / reconstruction / uniqueness

Predict changes without decoding again

Decode a picture from four sign patterns

Continuation: symmetry and missing information

A column swap exchanges 00 with 01 and 10 with 11. A row swap exchanges 00 with 10 and 01 with 11. These are permutations of the picture squares, not new sign conventions.

5. Choose a row swap or column swap. Predict exactly which coefficients stay the same and which change sign, then test your rule on (5,0,1,2). Explain using the cards themselves.

Before

5	0
1	2

My chosen swap

Coefficient prediction / why

6. An incomplete sensor reports only total brightness 8 and “top total minus bottom total” equal to 2. Find every nonnegative four-square picture consistent with it. Which further card readings make the picture unique?

Every allowed nonnegative picture / extra readings

Move the right endpoint

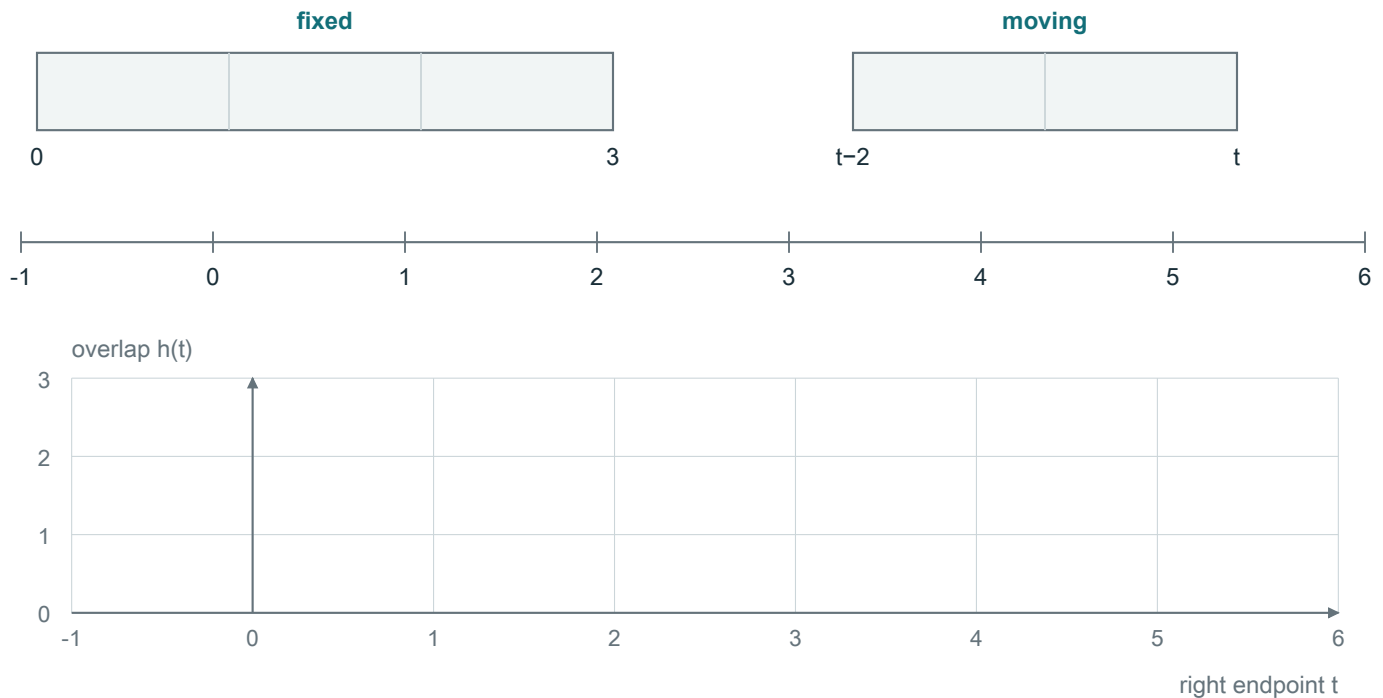
Predict what the sliding window draws

Core: lengths and prediction

The fixed window is $[0,3]$. The moving window is $[t-2,t]$, so t labels its right endpoint. Let $h(t)$ be the length covered by both windows. Touching at one point gives length zero. All lengths use the same unit.

1. Choose several positions t , including a first touch, a partly overlapping position and a fully nested position. Before making many measurements, sketch your prediction for the entire graph of $h(t)$. Mark where you think its rule changes.

Cut/trace the windows; 1 unit = 1 inch



2. Derive $h(t)$ for every real t . Explain each breakpoint using the two windows, including endpoint contacts. Use the formula to check and revise your prediction.

Endpoint cases / full formula / check every join

Build windows for a chosen graph

Predict what the sliding window draws

Algebra continuation: arbitrary positive widths

Now choose any lengths $a, b > 0$. The fixed interval is $[0, a]$ and the moving one is $[t-b, t]$. Its overlap function is $h_-(a, b)$. You may temporarily assume $a \geq b$ when describing cases.

3. Find the full formula for your chosen lengths, then for arbitrary $a \geq b > 0$. Swap the roles of a and b and prove whether the whole graph changes.

My widths / all endpoint cases / swapping the roles

4. Choose a desired distance $S > 0$ between the first and last contact positions, and a peak height $H > 0$, for such unit-height windows. Decide exactly when your design is possible and find every ordered pair (a, b) that realizes it. Test $S=7, H=2$.

My S and H / existence condition / every ordered pair

Why this is an integral operation

Predict what the sliding window draws

Continuation: a supplied integral interpretation

Define $f(x)=1$ on $[0,a]$ and 0 elsewhere, and $g(x)=1$ on $[0,b]$ and 0 elsewhere. Their convolution is $(f*g)(t)=\int_{-\infty}^{\infty} f(x)g(t-x)dx$. Supplied fact: the integral of an interval's indicator is its length; changing finitely many endpoint values does not change the integral.

5. Show exactly why this integral equals your overlap length. Explain what graph you would get if you instead slid $[t,t+b]$ while continuing to label its left endpoint t . Would making the input intervals open change the convolution?

Indicator inequalities / window location / endpoint convention

6. Find the area under your 3-by-2 overlap graph. Then prove the area under $h_{(a,b)}$ is ab for every positive a,b , using the graph's pieces. What does this area remember that the choice of fixed window does not?

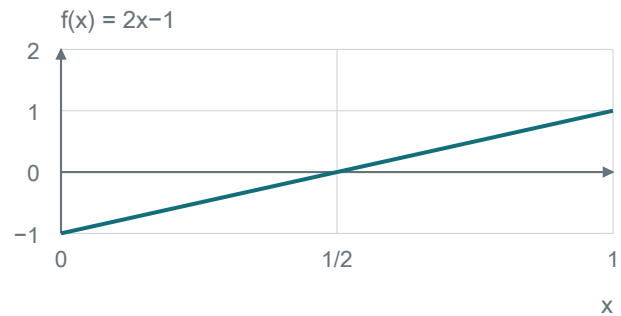
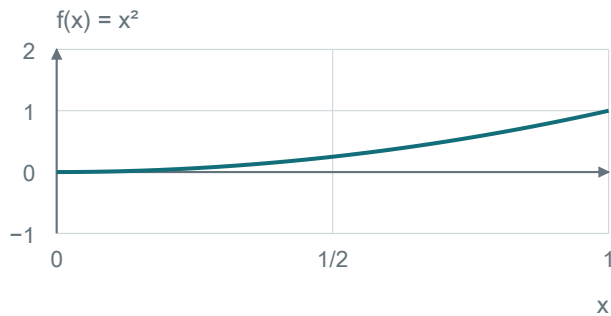
Area pieces / all positive a,b / information preserved

Choose a forcing and a dial

A graph feeds back its own mean

Core: definite integrals and a function equation

All functions are continuous and real-valued on $[0, 1]$. Seek $u(x) = f(x) + \lambda m$ for every x , where $m = \int_0^1 u(t) dt$ is u 's own mean. Choose $f(x) = x^2$ or $f(x) = 2x - 1$, and λ from $1/2, 1, 2$. Negative heights are allowed.



1. Choose a forcing and dial. Predict whether there is one solution, no solution, or more than one. Try a vertically shifted copy of the forcing graph and check whether its actual mean makes the feedback equation true.

My f and λ / predicted outcomes / candidate shift and actual mean

2. Introduce the unknown mean m before trying to determine every value $u(x)$. Derive an equation that every solution must satisfy. Use it to justify one impossible case from the available choices.

A necessary scalar equation / an impossibility certificate

Classify the whole equation

A graph feeds back its own mean

Core proof: necessity, construction and uniqueness

Keep the same equation but now let f be any continuous real function on $[0,1]$ and let λ be any real number. Write $F = \int_0^1 f(t) dt$.

3. Handle $\lambda=1$ completely. State exactly which f allow a solution and describe every solution when they exist. Prove both directions.

Critical case / necessity / construction / every solution

4. Handle every $\lambda \neq 1$. Give the solution, verify it directly and prove uniqueness. Put this case together with the critical case in a complete classification.

All other cases / direct verification / uniqueness

Repair the forcing, then perturb it

A graph feeds back its own mean

Continuation: exact sensitivity near the critical dial

You may adjust the forcing by a constant. First consider $f_a(x)=x^2-a$ at $\lambda=1$. Then move the dial to $\lambda=1-\delta$ with $0<\delta\leq 1$ and use $f_\varepsilon(x)=x^2-1/3+\varepsilon$.

5. Choose a so the critical equation has solutions. Describe all of them. If a is changed by any nonzero amount while λ stays exactly 1, what happens? Prove the claim.

Choose the adjustment / all solutions / a nonzero perturbation

6. For $\lambda=1-\delta$, compare the unique solution for ε with the one for $\varepsilon=0$. Find their maximum absolute difference. Given an allowed solution error $\eta>0$, how small must $|\varepsilon|$ be? Test $\varepsilon=1/100$ at $\delta=1/10$ and $\delta=1/100$.

ε	δ	solution difference
1/100	1/10	
1/100	1/100	

General difference / exact tolerance condition

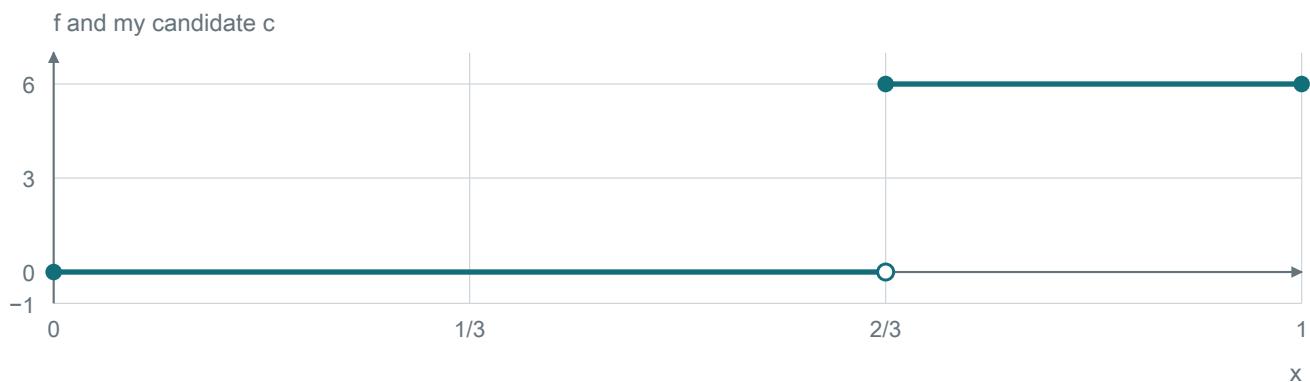
Choose how errors cost

Choose what best flat height means

Core: weighted areas and squares

The landscape is $f(x)=0$ for $0 \leq x < 2/3$ and $f(x)=6$ for $2/3 \leq x \leq 1$. A candidate is any real constant height c . Rule cards: $Q(c)=\int_0^1 (f-c)^2 dx$; $A(c)=\int_0^1 |f-c| dx$; $W(c)=\max_{0 \leq x \leq 1} |f(x)-c|$. For these step functions, each integral is a sum of rectangle widths times their heights.

1. Choose one rule card and two candidate heights. Predict the winner before calculating. Challenge your partner to improve the winner, and record which rule your comparison uses.



My rule / candidates / exact comparison

2. Using rule Q, find the best height among ALL real c . Give a certificate that excludes every better competitor, and decide whether any different height ties.

A bound for every real c / all equality cases

One landscape, different demands

Choose what best flat height means

Core proof: absolute values and worst error

Keep the landscape unchanged. A rule measures what it says: area under squared error, area under absolute error, or the single worst vertical error.

3. Find every best height for rule A. Your proof must include candidates below 0 and above 6, not just the heights you tried.

Absolute error: my cases, lower bound and every minimizer

4. Find every best height for rule W and prove the lower bound. Write a claim about “the best average height” that your three results refute, then repair the claim so it states a true mathematical question.

Worst error: proof / all minimizers / repaired claim

Design the widths

Choose what best flat height means

Continuation: a parameter and complete cases

Now f_p is 0 on $[0,p)$ and 6 on $[p,1]$, where $0 < p < 1$. Both heights occupy a positive width. Keep all three cost definitions.

5. Classify all minimizing heights for Q , A and W as p varies. Identify any value of p where a unique answer becomes a whole set of answers.

rule	minimizing height(s)	condition on p
Q		
A		
W		

Why these cases are complete

6. Choose a desired squared-loss optimum h with $0 < h < 6$. Build the low-region width that produces it and prove it is the only such width. Can that same landscape have h as its unique absolute-loss optimum?

My h / constructed width / uniqueness / absolute-loss comparison

Small height, chosen slope

Defeat a claimed derivative bound

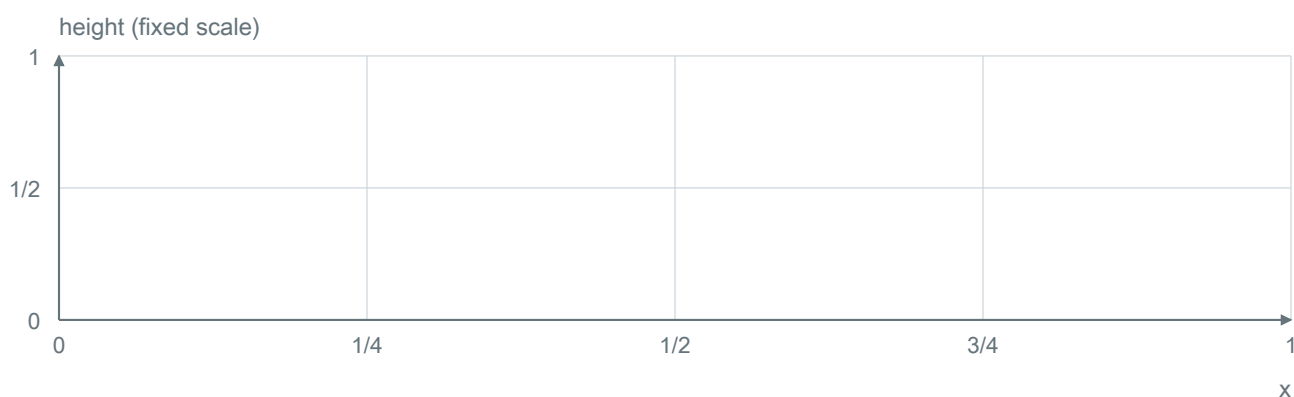
Calculus core: powers and derivatives

All graphs use $0 \leq x \leq 1$ and fixed axis scales. For a continuous real function, write $\|f\|_\infty = \max_{0 \leq x \leq 1} |f(x)|$. A polynomial is allowed; no graph may change its horizontal domain to win.

1. Choose a height allowance $\varepsilon > 0$ and a slope target $M > 0$. Build a polynomial $a x^n$ with $\|a x^n\|_\infty \leq \varepsilon$ and derivative at $x=1$ at least M . Give a rule that works for every chosen pair. Also meet $\varepsilon = 1/1000$ and $M=7$ exactly.

My ε, M / a, n / universal rule / supplied numerical target

2. For $f_n(x) = x^n/n$, $n=1,2,\dots$, find the exact norms of f_n and f_n' . Sketch two choices on the supplied common axes and explain which exact fact the sketch cannot certify on its own.



My two n values / exact norms / what the drawing cannot prove

Challenge every bound

Defeat a claimed derivative bound

Calculus proof: quantifiers and uniform convergence

A proposed bound is a finite $C \geq 0$ such that $\|f'\|_\infty \leq C\|f\|_\infty$ for EVERY continuously differentiable real function f on $[0, 1]$. Differentiation is called bounded under these norms if some such C exists.

3. Let a partner name C . Defeat the claim using a polynomial, then prove that your strategy defeats every finite C . Why is one unsuccessful candidate C not enough?

Partner's C / defeating polynomial / the quantifier argument

4. A graph sequence converges uniformly to zero when its maximum absolute height tends to zero. Do f_n and their derivatives both do so? Would $g_n(x) = x^n$ establish the same small-input claim? Explain the endpoint role.

Uniform heights / derivative endpoint / comparison with g_n

Change the measurement

Defeat a claimed derivative bound

Advanced continuation: norms and integrals

Now measure a C^1 input by $\|f\|^* = \|f\|_\infty + \|f'\|_\infty$, while still measuring its derivative by $\|f'\|_\infty$. Separately, define $If(x) = \int_0^x f(t)dt$ for continuous real f , using supremum norms for both input and output.

5. Find the smallest constant C for $\|f'\|_\infty \leq C\|f\|^*$. Does any nonzero function attain equality for this smallest C ? Distinguish a sharp bound from an attained ratio.

Sharp constant / possible equality / near-attaining inputs

6. Find the smallest constant C for $\|If\|_\infty \leq C\|f\|_\infty$. Prove the bound for every continuous f , and find an input attaining it. Explain what changed between this experiment and task 5.

Bound for all continuous inputs / an attaining input / comparison

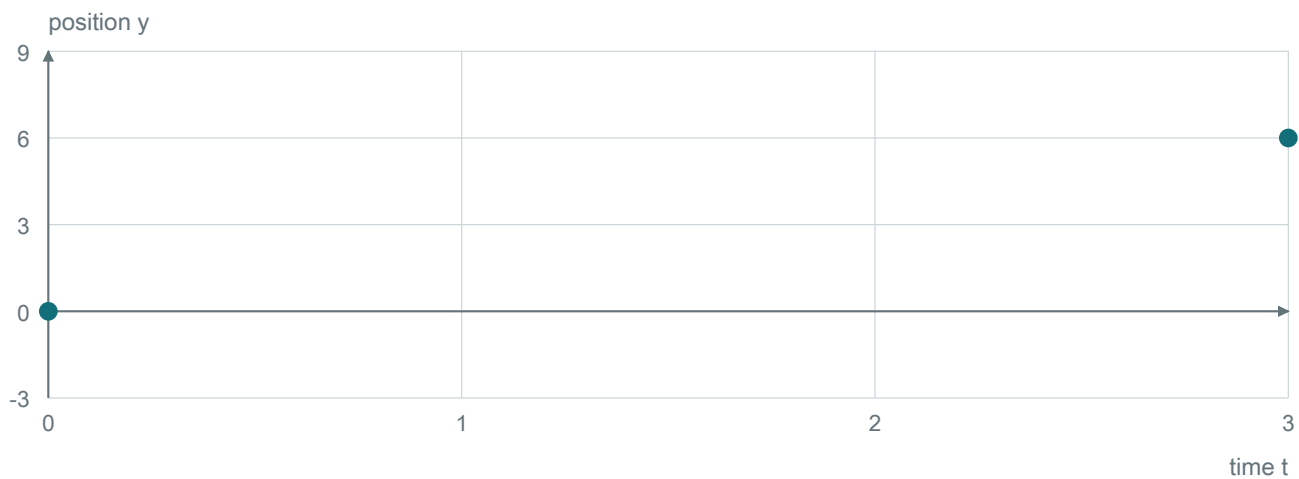
Invent a speed schedule

Choose a journey and its effort bill

Core: signed displacement and weighted squares

Start at $y=0$ and finish at $y=6$ after time 3. First travel at constant signed speed v for time 1, then w for time 2. The bill is $E=v^2+2w^2$. This invented quadratic-effort rule is not a physical fuel law.

1. Choose two different legal schedules, one with a pause or backward motion. Draw their position graphs and compare their exact bills. What equation decides whether a proposed schedule finishes correctly?



My two speed pairs / exact bills / legal endpoint equation

2. Find the cheapest legal two-speed schedule and prove uniqueness among all real v, w . Try expressing a rival schedule as a change from your candidate.

Candidate / excess-cost certificate / uniqueness

Let a rival use more changes

Choose a journey and its effort bill

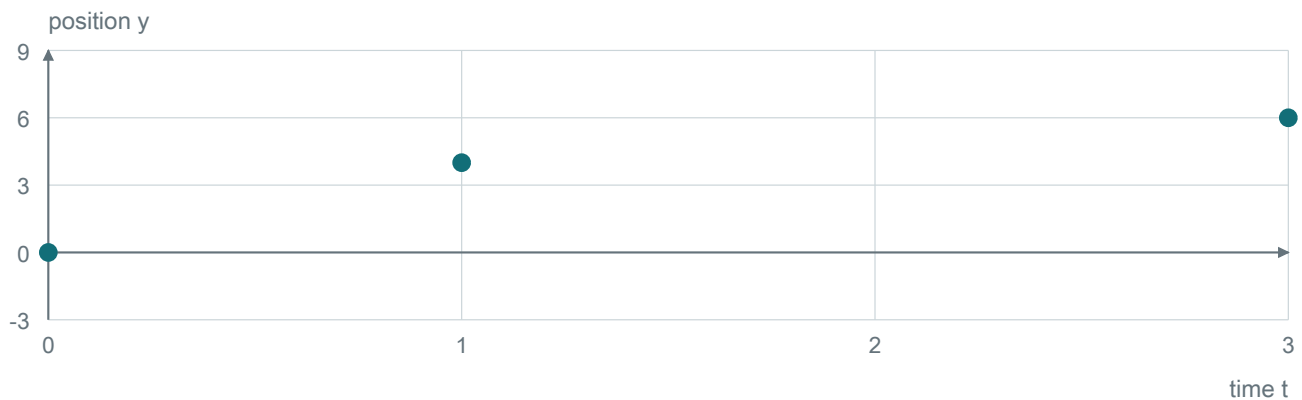
Core proof: finite weighted sums

A rival may choose any finite list of durations $\Delta t_i > 0$ summing to 3 and real speeds v_i satisfying $\sum \Delta t_i v_i = 6$. The bill is $E = \sum \Delta t_i v_i^2$. Changes of speed occur only at the finitely many joins.

3. Can any such rival beat your cheapest schedule? Derive a formula for its excess cost above 12 and say exactly when it is zero.

Any finite schedule / exact excess / equality conditions

4. Now require the trip to pass through $y=4$ at time 1. Find and prove the new cheapest finite schedule, allowing any extra speed changes before or after that time. Does it refute your earlier theorem?



New lower bound / attaining schedule / changed hypothesis

Every admissible curve

Choose a journey and its effort bill

Calculus continuation: FTC and equality in an integral

An admissible $y:[0,3]\rightarrow\mathbb{R}$ is continuous, is C^1 on each of finitely many closed subintervals, and satisfies $y(0)=0, y(3)=6$. One-sided derivatives at a join may differ; their point values do not affect $E=\int_0^3 (y')^2 dt$.

5. Prove the exact minimum and uniqueness for this whole class of curves, not only piecewise-linear schedules. Explain why the equality case determines the path at the joins as well.

Integral identity / equality on each piece / joins and endpoints

6. Test the family $y_a(t)=2t+a t(3-t)$ in your effort formula. Then change the bill to total distance $L=\int_0^3 |y'| dt$. Find its minimum over all admissible paths and all values of a for which this family attains it. Does uniqueness survive?

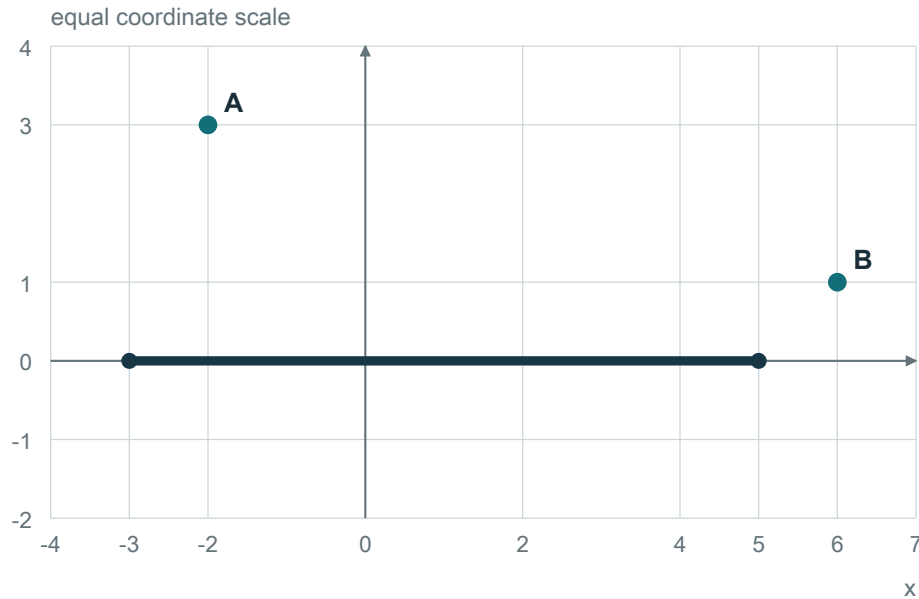
Effort of y_a / distance minimum / every attaining a / uniqueness

Place the contact point

A mirror that is too short

Core: a route, distances and a legal interval

$A=(-2,3)$, $B=(6,1)$. A legal route consists of the two straight segments $A \rightarrow M \rightarrow B$, where $M=(m,0)$ and $-3 \leq m \leq 5$. Minimize their total Euclidean length. Both endpoints lie above the mirror line.



1. Use string or drawn segments to choose and compare legal contact points. Try to improve your best route and record a conjectured winner. What would a proof need to exclude that your trials do not?

Chosen contacts / lengths / conjecture

2. Write the exact length $F(m)$. Is the midpoint of the mirror always a shortest contact? Test that claim on this diagram with an exact comparison.

Exact $F(m)$ / midpoint test / exact comparison

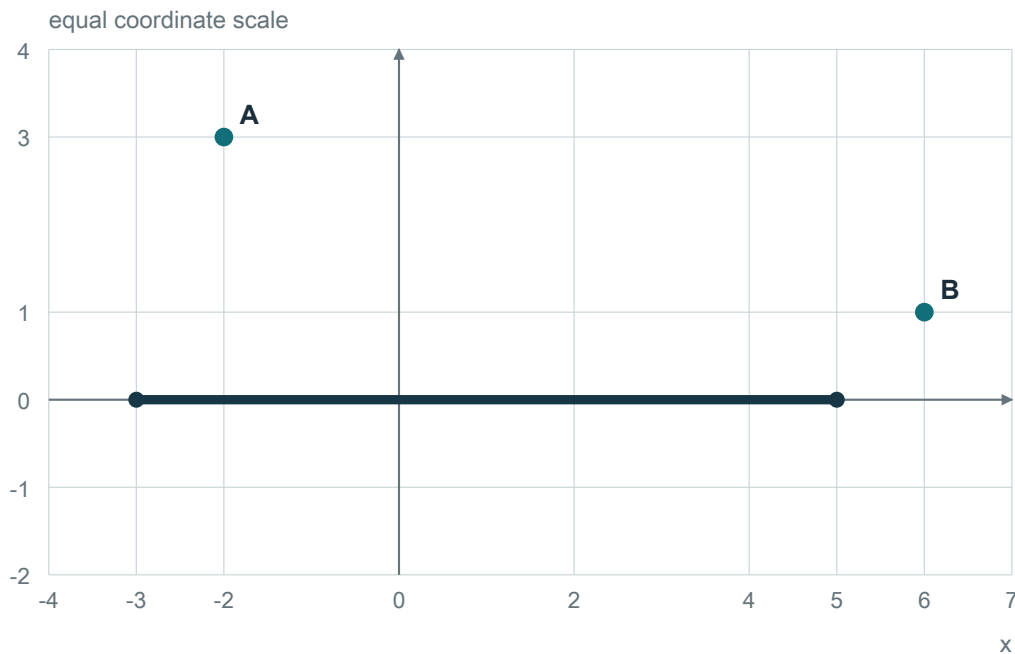
Unfold the route

A mirror that is too short

Core proof: reflection and equality

Reflect a point across $y=0$ by changing the sign of its y -coordinate. You may use the triangle inequality: $|UV|+|VW|\geq|UW|$, with equality exactly when V lies on the straight segment from U to W .

3. Reflect B to B' . Construct the crossing of the straight segment AB' with $y=0$ and find its exact coordinate. Is it a legal contact?



4. Prove the exact shortest length and the uniqueness of its contact. State where reflection and the equality condition each enter the argument.

Lower bound / when equality holds / why the contact is unique

Remove part of the mirror

A mirror that is too short

Proof continuation: a constrained minimum

Now only $-3 \leq m \leq 2$ is legal. Supplied tool: F is strictly convex, meaning $F((1-t)m+tn) < (1-t)F(m)+tF(n)$ for $m \neq n$ and $0 < t < 1$. Justification: apply the vector triangle inequality to each leg; vectors $(m+2, -3)$ and $(n+2, -3)$ cannot be positive parallel multiples unless $m=n$, and likewise for the other leg. Positive endpoint heights make the inequalities strict.

5. Choose the best contact on the shortened mirror. Compute its exact length. Explain why merely repeating the lower bound $4\sqrt{5}$ does not prove your new candidate is best.

Best legal contact / exact length / why the old bound is insufficient

6. Use the strict-convexity tool and the known unique minimum at 4 to prove that F strictly decreases to the left of 4 and strictly increases to its right. Finish the shortened-mirror proof, including uniqueness.

Strict decrease and increase / constrained optimum / uniqueness

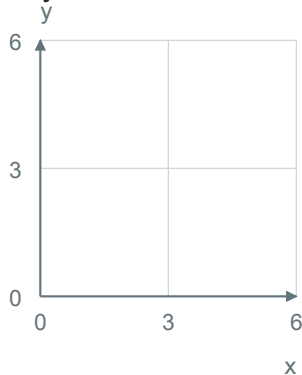
Challenge the splitter

Make a four-peg challenge

Core: two nonempty groups and their hulls

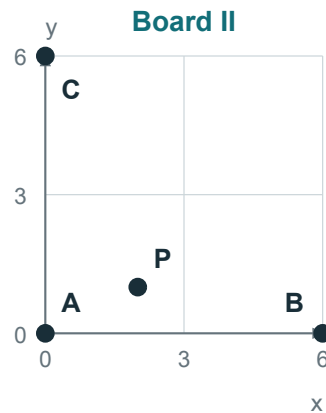
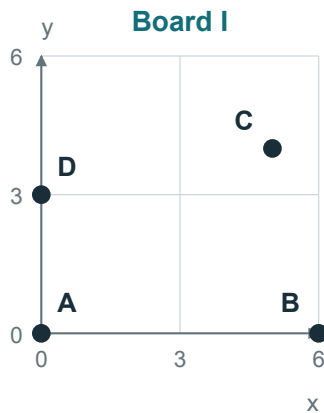
Use four distinct labeled points. Assign every label to exactly one of two nonempty groups. A group's convex hull is the smallest filled convex region containing it: one point stays a point, two make a segment, and three noncollinear points make a filled triangle. Boundary contact counts as an intersection.

1. Place four distinct pegs of your choice and trade the board with a partner. Find a legal split whose hulls meet, or try to explain why your arrangement defeats every split. Mark the actual common point when you succeed.



My point choices / split / actual contact

2. Solve both supplied boards: $A=(0,0), B=(6,0), C=(5,4), D=(0,3)$; and $A=(0,0), B=(6,0), C=(0,6), P=(2,1)$. Explain how the winning splits use different kinds of hulls.



The two splits and why their hulls meet

Can any arrangement defeat you?

Make a four-peg challenge

Core proof: all cases, including degeneracy

You may use this elementary planar hull fact: four distinct points that are not all collinear either include a point in or on the triangle of the other three, or are the four corners of a convex quadrilateral. Diagonals of a convex quadrilateral intersect inside it.

3. Prove that every four-distinct-point board has a legal split with meeting hulls. Include all-collinear points and three-collinear-plus-one-off-line points, and explain why your cases cover every board.

All board types / collinear cases / why the list is exhaustive

4. Allow different labels to occupy the same location. Does the conclusion change? Then show that three points are not enough to guarantee success for every board.

Coincident labels / a three-point counterexample / proof

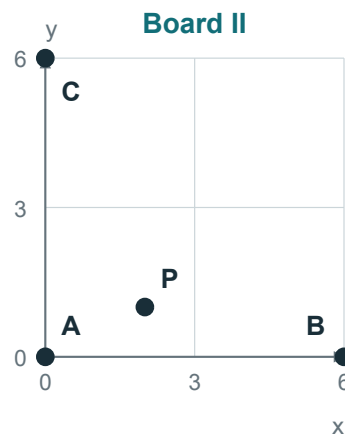
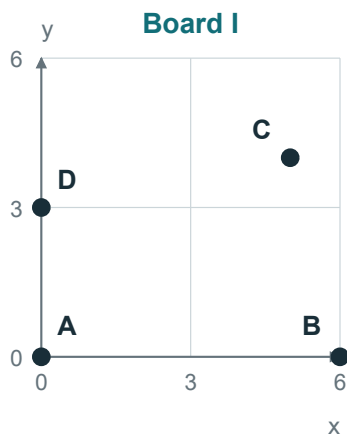
Certify contact with fractions

Make a four-peg challenge

Continuation: convex combinations and coordinates

A convex combination is $\sum w_i P_i$ with $w_i \geq 0$ and $\sum w_i = 1$. It belongs to the hull of those points. Use the same two coordinate boards as before; no diagram measurement counts as an exact fraction.

5. Find the exact crossing of AC and BD in the first board. Express it as a convex combination from each group, and explain why the coefficients certify actual segment contact.



Board I: exact crossing / two nonnegative weight systems

6. Express P from the second board as a convex combination of A,B,C. Move all terms to one side to obtain a signed relation with coefficient sum zero. Why do signed coefficients alone not describe a point inside one hull?

Board II: weights / signed relation / a signed-coefficient obstruction

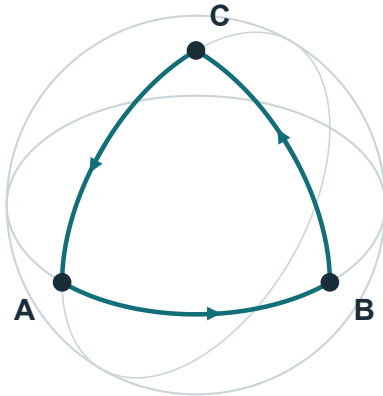
Carry an arrow around the corner

Let the route turn a tangent arrow

Core: spatial directions and an exact rule

Use the unit sphere with $A=(1,0,0)$, $B=(0,1,0)$, $C=(0,0,1)$. Route $A \rightarrow B \rightarrow C \rightarrow A$ follows the short great-circle arcs. On one leg let T be its forward unit tangent and N the fixed unit normal to that leg's oriented plane. Write the carried arrow as $aT+bN$ and keep a,b constant along that leg. At a corner keep the actual arrow fixed, then use the NEXT leg's T,N . Do not turn it with the walker.

1. Choose initial arrow $+y$, $+z$ or a mixture in A 's tangent plane. Predict its final direction, then try the route on the ball. Record the arrow at each landmark and any disagreement between prediction and experiment.



Arrows show route direction only.

My initial arrow / prediction / vectors at B,C,A

2. Starting from the same chosen arrow, compare $A \rightarrow B \rightarrow A$ with $A \rightarrow C \rightarrow B \rightarrow A$. Predict whether each restores the arrow. Explain why simply turning the arrow with the walker would be a different rule.

Retraced and reversed routes / my experiment / rule distinction

Track the two basis arrows

Let the route turn a tangent arrow

Core proof: orthonormal frames and signed components

For $0 \leq t \leq \pi/2$ the supplied frames are: AB has $r=(\cos t, \sin t, 0)$, $T=(-\sin t, \cos t, 0)$, $N=+z$; BC has $r=(0, \cos t, \sin t)$, $T=(0, -\sin t, \cos t)$, $N=+x$; CA has $r=(\sin t, 0, \cos t)$, $T=(\cos t, 0, -\sin t)$, $N=+y$. In each case T, N are perpendicular unit vectors tangent at r .

3. Use the frame cards to compute the complete route first for initial $+y$ and then for initial $+z$. At each corner explicitly rewrite the arriving vector in the next frame.

initial arrow	at B / next-frame form	at C / next-frame form	back at A
$+y$			
$+z$			

4. Find the final vector for every initial $a \cdot y + b \cdot z$. Prove the map preserves length and identify its signed angle in A's tangent plane, where positive means from $+y$ toward $+z$. Justify the inverse and retracing claims.

Every a, b / length / signed angle / inverse and retracing proofs

Choose a different longitude

Let the route turn a tangent arrow

Trigonometric continuation: changing frames

Keep $A=(1,0,0)$, $C=(0,0,1)$, but choose $B=(\cos \alpha, \sin \alpha, 0)$, $0 < \alpha < \pi$. Travel the equator $A \rightarrow B$ and the short meridians $B \rightarrow C \rightarrow A$. Put $c = \cos \alpha$, $s = \sin \alpha$. Endpoint frames: AB has $T: +y \rightarrow (-s, c, 0)$, $N = +z$; BC has $T: +z \rightarrow (-c, -s, 0)$, $N = (s, -c, 0)$; CA has $T: +x \rightarrow -z$, $N = +y$.

5. Derive the returning vector for initial $+y$ and $+z$ using these frames. Then give the formula for an arbitrary initial $a \cdot y + b \cdot z$. Do not assume an angle-area theorem.

Transport $+y$ / transport $+z$ / combine for arbitrary a, b

6. Choose a longitude separation and an initial vector; compute the final vector and test its length. Include the case $\alpha = \pi/3$, initial $+y$. Can a nonzero vector be unchanged after one trip when $0 < \alpha < \pi$?

My α and vector / $\pi/3$ test / norm / fixed-vector obstruction

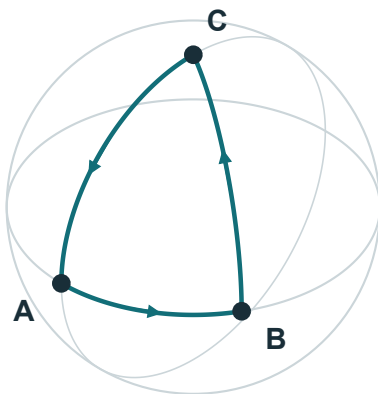
What did the route measure?

Let the route turn a tangent arrow

Continuation: area and limits of a model

The unit sphere has area 4π . A lune of longitude width α occupies fraction $\alpha/(2\pi)$ of the sphere; its northern half is the triangle $A \rightarrow B \rightarrow C \rightarrow A$. Angles are measured in radians.

7. Compute the triangle's area and compare it with the rotation you proved. Why is this a proved relation for this family of routes, but not a proof about all closed curves on every surface?



schematic: longitude α

Lune area / northern half / proved rotation / scope

8. A partner proposes carrying an arrow fixed in surrounding three-dimensional space. Test a fixed $+\gamma$ arrow along AB. Give a closed route that does restore every arrow, and explain why curvature does not force every closed route to turn it.

Tangency test / a restoring route / what curvature alone does not force

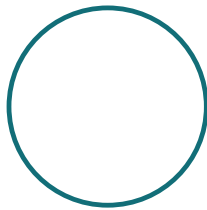
Choose the point to remove

Test a reversible reshaping

Core: trace pieces without jumping gaps

The pictures represent every point along the drawn strands, not just marked vertices. “Delete a point” means that single mathematical point is unavailable. A connected piece can be traced between any of its retained points in these examples. The figure-eight crossing and the Y center are joined; the two separate loops are not joined.

1. Choose points to delete from the circle and the closed segment. Try different positions and record the number of pieces left. Can a segment endpoint trick a test based on just one careless choice?

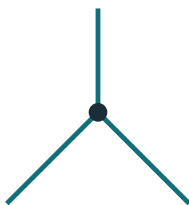


circle



closed segment

2. Choose useful punctures in the Y, the joined figure-eight and the two disjoint loops. Classify the possible numbers of remaining pieces for each shape. Explain the role of a junction instead of treating every drawn point alike.



Y: center joined

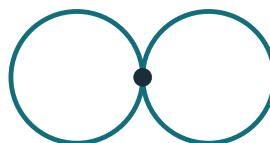


figure-eight: joined



two separate loops

My punctures / remaining pieces / all possible cases

Turn the test into a proof

Test a reversible reshaping

Topology proof: supplied facts and matching points

A homeomorphism is a continuous bijection with continuous inverse. Supplied facts: an interval is connected, and a continuous image of a connected space is connected. A homeomorphism restricts to a homeomorphism after deleting matching points, because the map and inverse remain continuous on their remaining subspaces.

3. Prove that the circle cannot be homeomorphic to $[0,1]$. Choose a point strategically, and justify why every corresponding punctured circle is connected.

Strategic point / matched puncture / connectedness argument

4. Use matching-point deletion to distinguish the Y from a segment and the joined figure-eight from a circle. Also distinguish the joined figure-eight from two disjoint circles.

Y / figure-eight / two loops: the invariant and contradiction

Test actual maps

Test a reversible reshaping

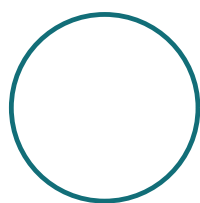
Continuation: bijections, inverses and coordinates

A continuous map onto a shape is not automatically a reversible reshaping. Test which of the definition's requirements each candidate satisfies.

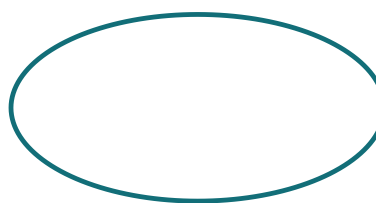
5. The map $(x,y) \mapsto x$ sends the unit circle onto $[-1,1]$. The map $t \mapsto (\cos 2\pi t, \sin 2\pi t)$ sends $[0,1]$ onto the circle. Why does neither contradict your impossibility proof? Exhibit the failure explicitly.

Candidate maps / repeated images / failed homeomorphism condition

6. Build a homeomorphism from the unit circle to the ellipse $(X/2)^2 + Y^2 = 1$. Give its inverse, check both compositions and explain continuity. What geometric features did this successful map change?



unit circle



ellipse: horizontal semiaxis 2

Both pictures use the same coordinate scale.

My map / inverse / both compositions / continuity

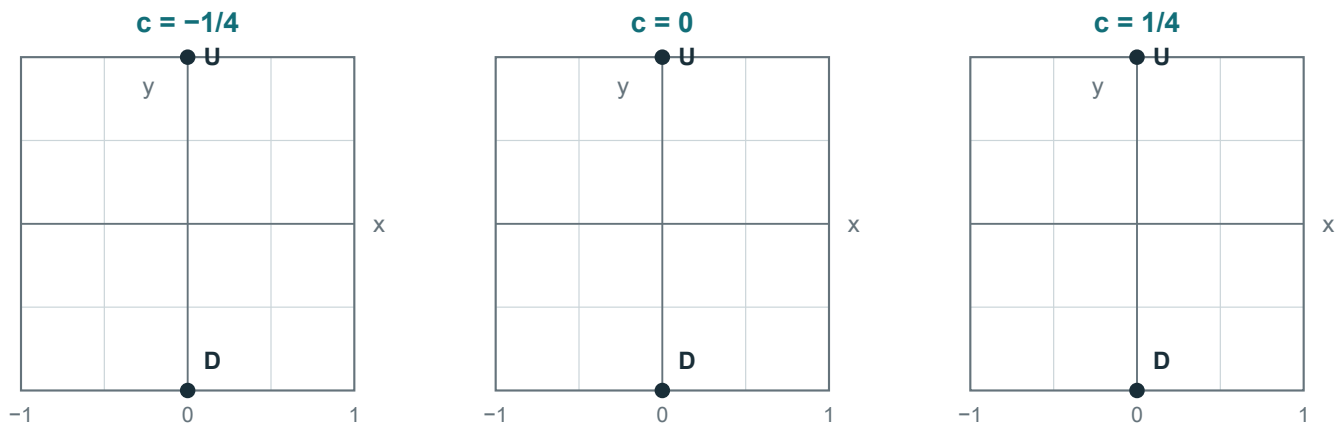
Choose a water level and a route

Find the flood's joining height

Core: squares, inequalities and wet paths

Work only in the square $-1 \leq x \leq 1$, $-1 \leq y \leq 1$. Ground height is $f(x,y)=x^2-y^2$. At level c , a point is wet exactly when $f \leq c$; equality is included. A wet route is a continuous path whose every point is wet.

1. Choose one negative and one positive water level and test points of your own. Then sketch the exact wet regions at $c=-1/4, 0, 1/4$ on the supplied grids, labeling their boundary inequalities.



My extra levels / tested points / boundary inequalities

2. Try to connect $U=(0,1)$ to $D=(0,-1)$ by a wet route at each common level. Give an explicit successful route or an obstruction that rules out every route. At which level do the two sides first join?

Common levels: routes or all-route obstructions / first joining

Classify every flood level

Find the flood's joining height

Core proof: bounds, connectedness and a barrier

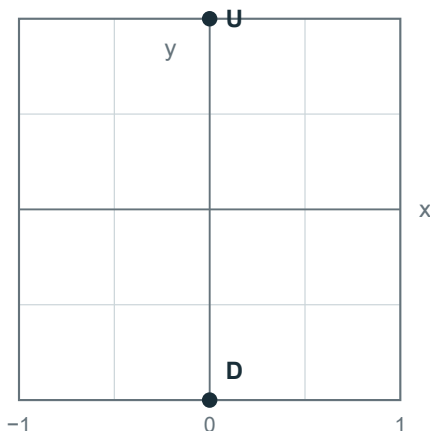
The domain remains the closed square. A component is a maximal connected wet piece. You may use that any continuous route from positive y to negative y passes through $y=0$.

3. Classify the number of wet components for every real c , including the first nonempty level, the joining level and when the whole square is wet. Prove that each claimed piece really is connected within the square.

All real levels / endpoints / paths within every claimed component

4. Choose a forced crossing point $(a,0)$, with $|a| \leq 1$. Among all U-to-D routes that pass through it, find the smallest water level making an entire route wet. Give both a lower bound and a route attaining it, including $a=0$ and $a=\pm 1$.

My forced crossing



Chosen a / lower bound / legal attaining route

What makes this a Morse saddle?

Find the flood's joining height

Calculus continuation: partial derivatives and Hessian

A critical point has zero gradient. The Hessian is the matrix of second partial derivatives. A critical point is nondegenerate if its Hessian is invertible; its index is the number of negative eigenvalues. These derivative tests concern the function near an interior point.

5. Find all critical points and the Hessian of x^2-y^2 . Compare with the bowl x^2+y^2 near zero: what different component event occurs as c crosses 0?

function	critical point(s) / Hessian	event near zero
x^2-y^2		
x^2+y^2		

Nondegeneracy / index / justification of the component event

6. The landscape $g=x^4-y^4$ has the same wet set as f at level 0. Verify that statement, then test whether g has a nondegenerate critical point at the origin. Why can a matching drawing not justify invoking the Morse lemma?

Exact zero-level comparison / gradient and Hessian / theorem hypothesis

Hide a number

How much can one question guarantee?

Core: fractions and comparisons

An unknown real number r lies strictly between 1 and 2. Each test asks which side of your chosen number contains r ; an exact hit wins. Your partner may delay choosing r , but every reply must stay consistent with all earlier replies. Keep the entire interval still possible. Your strategy must handle unlucky replies too.

1. Choose a first test. Your partner gives the reply leaving the wider possible interval. Continue for four questions, always choosing your next test. Record the surviving interval after each reply.

Test 1: _____

surviving endpoints: _____

Test 2: _____

surviving endpoints: _____

Test 3: _____

surviving endpoints: _____

Test 4: _____

surviving endpoints: _____

2. Find a strategy that guarantees width at most $1/16$ after four questions. Prove that no comparison strategy can guarantee a smaller width for every secret number.

Your construction / explanation

A bracket for a root

How much can one question guarantee?

Core continuation: squares; a supplied continuity theorem

Now the target is the positive solution of $x^2=2$. A bracket $[a,b]$ has $a^2\leq 2\leq b^2$ and $1\leq a\leq b\leq 2$. Supplied theorem: a continuous function taking opposite signs at two endpoints has a zero between them. Polynomials are continuous.

3. Start with $[1,2]$. Make four midpoint tests, recording each exact square and the bracket you keep. Give one estimate and a proved bound on its error.

Test	Exact square	Kept bracket
1		
2		
3		
4		

4. Before doing more arithmetic, how many midpoint tests guarantee bracket width below $1/100$? Would opposite signs still certify a root for a function allowed to jump? Make a counterexample.

Your construction / explanation

A tangent can make no progress

How much can one question guarantee?

Calculus continuation: derivatives

Newton's rule is $x_{\text{next}} = x - f(x)/f'(x)$, whenever $f'(x) \neq 0$. Use $f(x) = x^3 - 2x + 2$. This page concerns the rule's guarantee, not a race to many decimal places.

5. Try Newton's rule from $x=0$ and from $x=1$. Can this rule continue forever without approaching any root? Explain exactly what happens.

Start / next values / exact explanation

$x = 0$

$x = 1$

6. Find a root bracket of width $1/4$ inside $[-2, -1]$, using two midpoint tests. Explain why this certificate remains valid even though Newton's rule cycled elsewhere.



Signs / retained bracket / guarantee

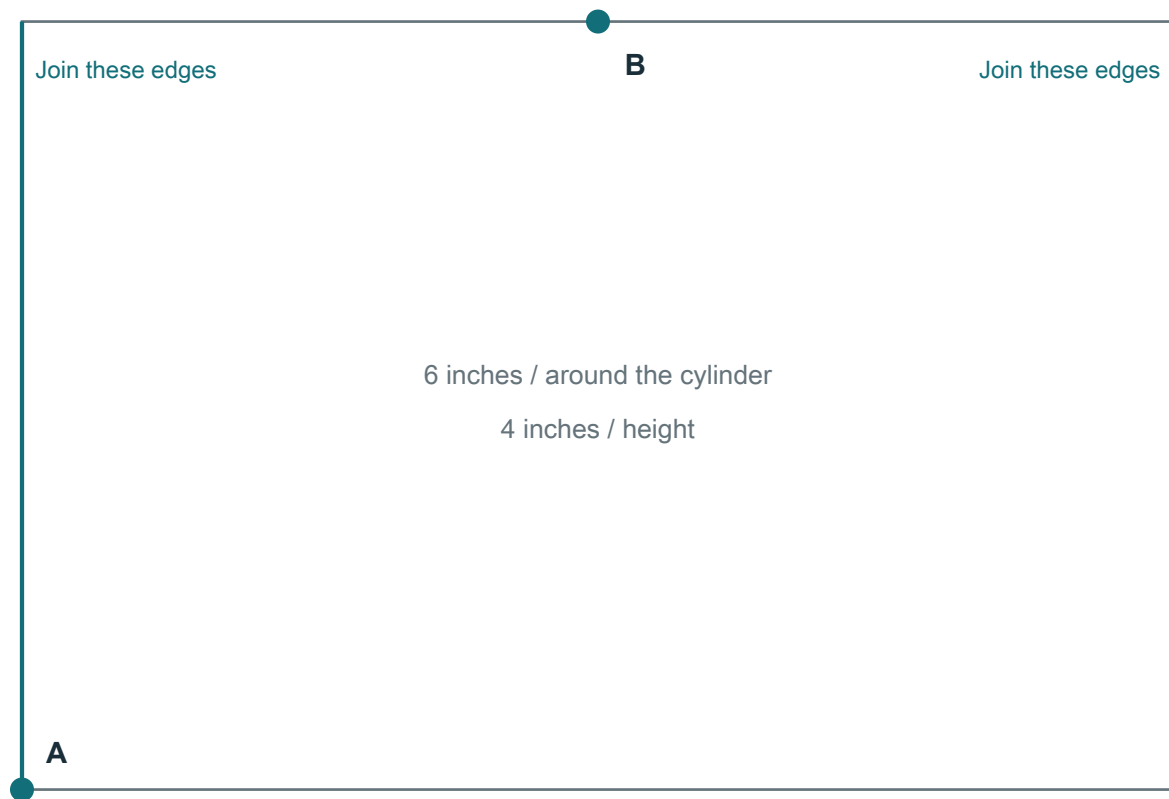
A route the seam hides

Find every shortest route around a cylinder

Core: paper construction, length comparison and the Pythagorean theorem

Cut out the rectangle from a spare copy. Tape the two short edges of the 6-by-4 rectangle edge-to-edge, without overlapping the paper, to make a cylinder. A is on the bottom rim at the seam; B is on the top rim halfway around. Routes must stay on the curved side, including its rims. No end caps or shortcuts through the interior.

1. Use string to find two different routes from A to B. Choose a cut that makes one route easy to draw flat. Can a different cut reveal a route you first missed?



2. Draw a route that becomes straight when unrolled. Calculate its length, then find a different route with the same length. Can you yet rule out every shorter route?

Length / second route / what still needs proof

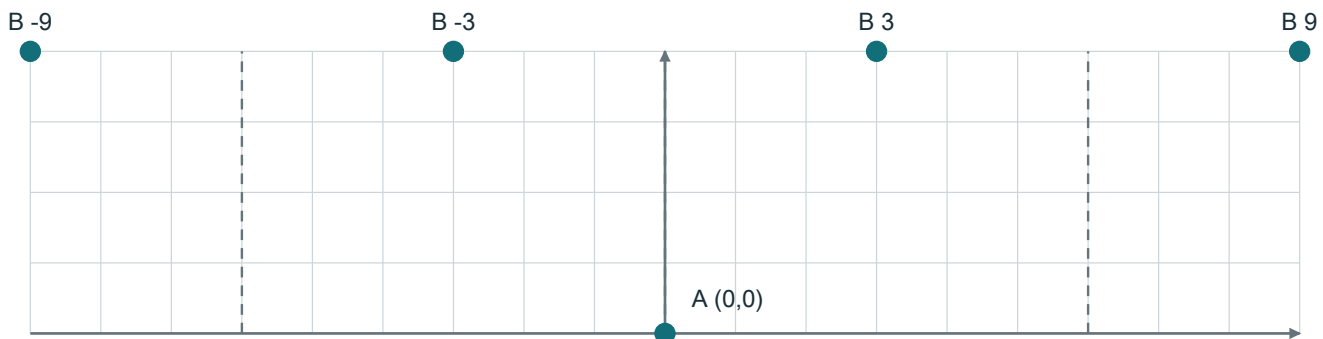
A map with many copies

Find every shortest route around a cylinder

Core continuation: integer shifts and a lower bound

The repeated strip has $A=(0,0)$. Copies of B occur at $(3+6k,4)$, one for every integer k . Any surface path starting at A can be followed continuously across adjacent copies of the rectangle without jumping.

3. Choose three copies of B, including one farther away. Draw a straight route to each, calculate its length, and roll or trace how the extra winding appears.



Copies continue in both directions; height is always 4.

Chosen copies / lengths

4. Prove that there are exactly two shortest routes. Your argument must include curved routes and arbitrarily many windings, not just the copies printed on this page.

Your construction / explanation

Move the destination

Find every shortest route around a cylinder

Continuation: piecewise formulas and comparison

Keep A at the bottom seam. Move B around the top rim: its horizontal offset is d with $0 \leq d < 6$, and its height stays 4. Its copies are $(d+6k, 4)$.

5. Choose offsets on both sides of $d=3$. Find the winning lift for each. Locate exactly where the winner changes and where there is a tie.



Choose d and mark the winning horizontal displacement.

Comparisons / switch / tie

6. A long helical route looks perfectly straight in every small unrolled patch. Does that make it globally shortest? Give a specific counterexample from your map.

Your construction / explanation

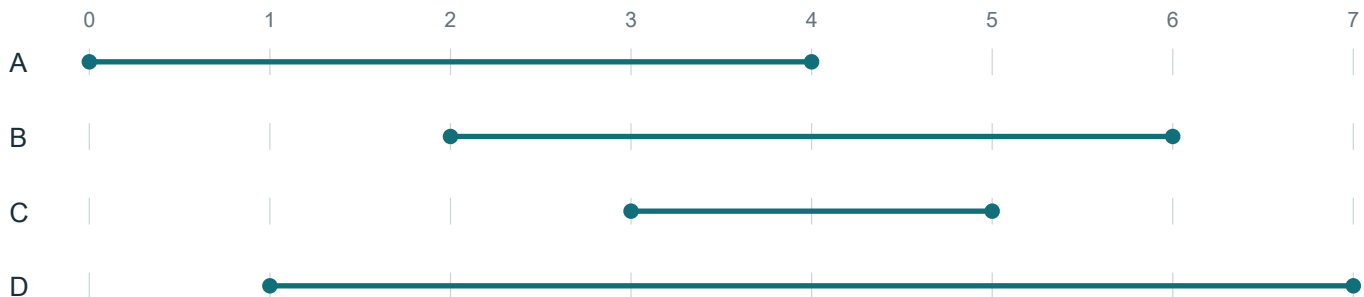
Try to defeat the group

Can every pair agree and everyone disagree?

Core: closed intervals on a line

Each person accepts every point in one closed interval $[left, right]$, including endpoints. A pair agrees if their intervals overlap. The whole group agrees only if one point belongs to every interval.

1. Find all common choices for $[0,4]$, $[2,6]$, $[3,5]$ and $[1,7]$. Then choose four new intervals: try to make every pair agree while the whole group disagrees.



Common part / your attempted counterexample

2. A shortcut checks that everyone agrees with one favorite interval. Make three intervals that pass this shortcut but have no common choice.

Draw and label your chosen intervals

The smallest certificate

Can every pair agree and everyone disagree?

Core continuation: an extremal argument

Use any finite list of nonempty closed bounded intervals. Endpoints need not be integers. You may point to a few decisive intervals instead of checking every pair.

3. Explain how to find the entire common interval from just two endpoint extremes. If the common part is empty, show that two of the original intervals already prove failure.

Your construction / explanation

4. Design a family whose only common point is 3. Then explain why sharing a single endpoint counts as agreement and why your proof works even if the two decisive extremes come from the same interval.

Draw and label your chosen intervals

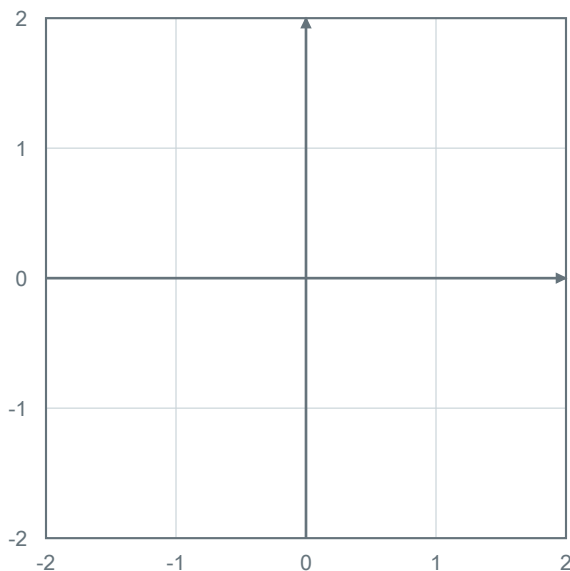
Change the allowed shapes

Can every pair agree and everyone disagree?

Continuation: signed coordinates and inequalities

In the square $-2 \leq x \leq 2$, $-2 \leq y \leq 2$, let A allow $x \geq 0$, B allow $y \geq 0$, and C allow $x+y \leq -1$. Each allowed region is convex: the segment between two allowed points stays allowed.

5. Shade the three regions. Mark a witness for each pair's agreement. Prove there is no point accepted by all three. Which feature of the earlier setting changed?



Pair witnesses / no common point

6. Return to a line but permit gaps: allowed sets are $\{0,1\}$, $\{1,2\}$ and $\{0,2\}$. Check every pair and the whole group. State exactly what your interval theorem does and does not promise.

A



B



C



Pair intersections / triple intersection / exact scope

Two bands, two predictions

Predict the cut before using scissors

Core: construction and tracing

Band P joins the ends of a strip without a turn. Band M joins them after one half-turn. Draw the center line before taping. A boundary loop follows an actual edge until it first returns to its starting point.

1. Trace every boundary loop of each band. Then predict how many separate pieces and how many boundary loops each piece will have after a cut exactly along the center line. Record before cutting.

Band	Before: boundary loops	After: pieces / loops per piece
P		
M		

2. Cut, trace and compare. Can one connected piece have two boundary loops? Give evidence that distinguishes “pieces” from “boundary loops.”

Your construction / explanation

Explain on a flat strip

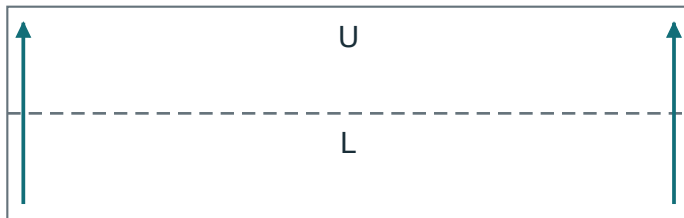
Predict the cut before using scissors

Core continuation: follow a seam rule

Name the upper half U and lower half L before the center cut. Plain gluing matches upper to upper and lower to lower. Half-turn gluing matches upper to lower and lower to upper. The cut separates U and L along their long middle edge, but their end identifications remain.

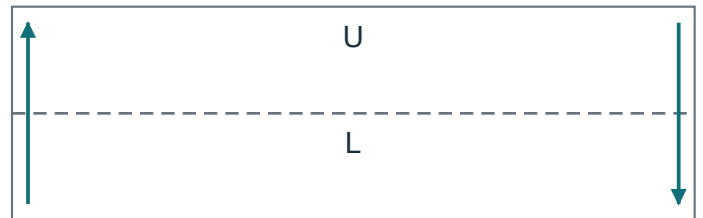
3. Draw the seam journeys after cutting P and M. Use these journeys to prove the number of connected pieces without relying on the photograph or physical result.

P / matching ends



Seam journey / pieces

M / half-turn ends



Seam journey / pieces

4. For the cut Möbius band, explain why the connected result has matching ends when viewed as one longer strip, and why it has two boundary loops. An ordinary band with matching ends is called an annulus.

Your construction / explanation

A different cut to predict

Predict the cut before using scissors

Continuation: three lanes

Make a fresh half-turn band with two parallel cut lines dividing its width into three equal lanes: upper U, center C and lower L. Predict first, then cut both lines completely. At the seam, U joins L, C joins C and L joins U.

5. Predict the connected pieces. For each, decide whether the result is an ordinary band (annulus) or a Möbius band. Track lane names and whether a small arrow across the strip's width reverses when a journey closes.

U	L
C	C
L	U

Labels on the right show which left lane the seam meets.

Your journeys / surface types / boundary loops

6. Invent a claim that confuses linked pieces, connected pieces or boundary loops. Give your partner a seam-based correction using one of the bands you made.

Your construction / explanation

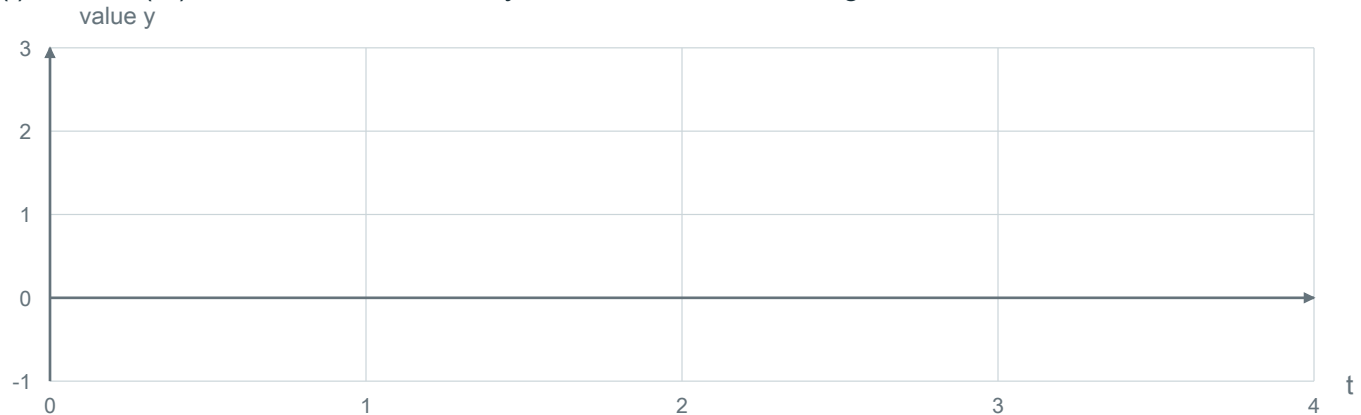
Two starts, one rule

Where did the starting value go?

Core: derivatives and exponential functions

A system follows $y' = 1 - y$ for $t \geq 0$. Choose a starting value a from -1 , 0 or 2 . Compare with a partner who chose a different start. No transform rule has been assumed yet.

1. Predict the initial slope and which level the system moves toward. Choose a candidate of the form $y(t) = 1 + C e^{-t}$. Determine C and verify the rate rule and starting value.



Chosen start / candidate / verification

2. A shortcut claims that every starting value gives $Y(s) = 1/[s(s+1)]$ and hence $y(t) = 1 - e^{-t}$. Find an exact check that exposes the mistake. What information must an algebraic shortcut preserve?

Your construction / explanation

Recover the missing term

Where did the starting value go?

Core continuation: improper integrals and integration by parts

Define $Y(s) = \int_0^\infty e^{-st} y(t) dt$. Work with real $s > 0$ for these examples. Supplied table: $L(1) = 1/s$, $L(e^{-t}) = 1/(s+1)$, $L(t e^{-t}) = 1/(s+1)^2$. Linearity follows from integration. Do not assume the derivative rule.

3. Integrate $e^{-st} y'(t)$ by parts first from 0 to T . Then let T increase. For your candidate from page 1, justify the boundary limit and derive $L(y')$. Where does a appear?

Finite endpoint T first; then justify $T \rightarrow \infty$

At T : _____

At 0: _____

4. Transform $y' + y = 1$ while keeping a visible. Recover $y(t)$, then prove that two differentiable solutions with the same initial value must agree.

Your construction / explanation

Change the input

Where did the starting value go?

Continuation: transform algebra and a maximum

Now $y' + y = e^{-t}$, $y(0) = 0$. The same transform table applies. First sketch what a fading input might do to the output, then calculate.

5. Find the transformed equation and a candidate $y(t)$. Verify both the rate equation and the initial value directly. Does this output keep increasing forever?



Transform / candidate / verification

6. At the peak, compare the input e^{-t} with the loss $y(t)$. Explain the peak using the original rate rule, and show why a transform table alone would not explain the missing starting-value term.

Your construction / explanation

Choose a target height

Infinitely many doublings before the deadline

Core: rational functions, derivatives and limits

A quantity obeys $y'=y^2$ and starts at $y(0)=1$. A proposed trajectory is $y(t)=1/(1-t)$, initially for $0\leq t<1$. The formula is a candidate to check, not a graph-based guess.

1. Choose three target heights greater than 1. Find their arrival times from the candidate formula. Include a very large target, and decide whether any finite target reaches time 1.

Chosen height H	Arrival time	Why it is before 1

2. Verify the rate rule and starting value. Find the times for heights $2,4,8,\dots,2^n$ and the gaps between consecutive doublings. What do their total times approach?



Mark arrival times for your chosen powers of 2.

Derivative check / nth arrival / gaps

Is the far branch a rescue?

Infinitely many doublings before the deadline

Core continuation: continuity and uniqueness

The same expression $1/(1-t)$ is also defined for $t > 1$. A continuation must be one real-valued differentiable function on a time interval containing the original trajectory and the later times. Supplied local theorem: a continuously differentiable rate law in y gives local uniqueness.

3. Check the formula's rate rule on $t > 1$. Then decide whether that branch continues the solution starting at 1. Prove your answer using the behavior near time 1.

Before time 1

t = 1

After time 1

t = 1

What a real-valued continuation would require

4. Someone says, "The rate y^2 is finite at every finite y , so the solution must exist forever." Diagnose the claim. How can local uniqueness and finite-time escape both hold?

Your construction / explanation

Change the start

Infinitely many doublings before the deadline

Continuation: classify positive, zero and negative starts

Use $y(0)=a$ and the candidate $y_a(t)=a/(1-at)$, wherever its denominator is nonzero. Consider forward time $t\geq 0$ only.

5. Choose one positive, one zero and one negative a . Verify the formula, find each maximal forward interval and describe its endpoint behavior. Then give the rule for every real a .

Your start a	Forward interval	Long-time / endpoint behavior
Positive:		
Zero:		
Negative:		

6. For a positive solution, differentiate $z=1/y$. Use its simple rule to recover the blow-up time. Compare with $y'=y$ from the same positive start: does every unbounded solution blow up in finite time?

Your construction / explanation

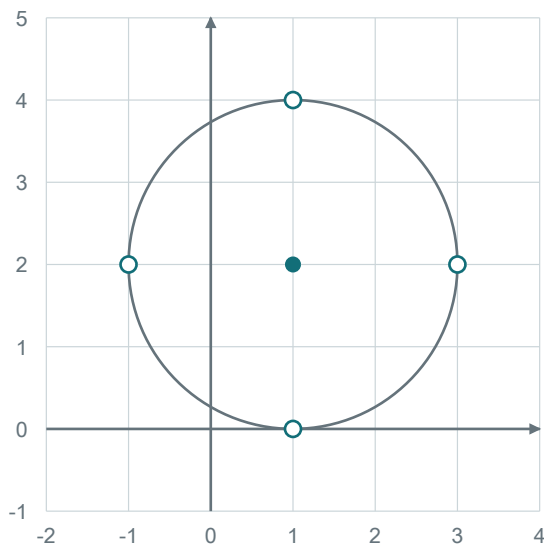
Move the sampling cross

Can four samples fool the average?

Core: coordinates and signed squares

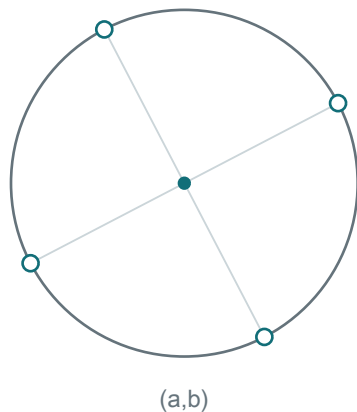
Let $h(x,y)=x^2-y^2$. Start with a circle centered at $(1,2)$ with radius 2. Use the four cardinal points, then choose a different center or radius. The average of four samples means their sum divided by 4.

1. Mark the four points and evaluate h there. Compare their average with h at the center. Choose a second circle and try to break the pattern.



Values / average / your second circle

2. Rotate the four-point cross. Write its displacement vectors as $(u,v), (-v,u), (-u,-v), (v,-u)$, where $u^2+v^2=r^2$. Prove the average is still the center value for every center and rotation.



Displacements / cancellation / exact identity

Design a test, then fool it

Can four samples fool the average?

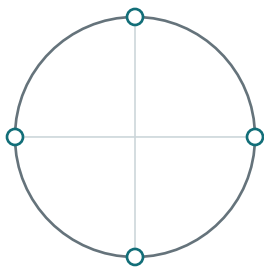
Core continuation: quadratic algebra

Now $q(x,y)=A x^2+B xy+C y^2+D x+E y+K$. Use four quarter-turn-spaced samples on any circle, with any center and starting angle.

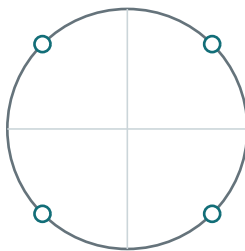
3. Which coefficients make every such four-sample average equal q at the center? Find a necessary and sufficient condition, not just examples.

Your construction / explanation

4. The function $p(x,y)=x^2y^2$ passes the cardinal-point test on the unit circle centered at 0. Rotate the sample cross by 45° . Does it still pass? Explain what the first test did not prove.



Cardinal



Rotated 45°

Sample values

What the first test did not prove

Average the whole circle

Can four samples fool the average?

Calculus continuation: trigonometric integrals

The continuous circle mean is $(1/(2\pi)) \int_0^{2\pi} q(a+r \cos \theta, b+r \sin \theta) d\theta$. It averages every angle equally. This is different from checking four points.

5. Compute this mean for a general quadratic q and compare with your finite classification. Then calculate $p=x^2y^2$ on a radius- r circle centered at the origin.

Your construction / explanation

6. Define the Laplacian $\Delta q = q_{xx} + q_{yy}$. Relate it to the coefficient condition. State precisely which statements you proved, and which general claim about harmonic functions would need a further theorem.

Your construction / explanation

Build a feedback experiment

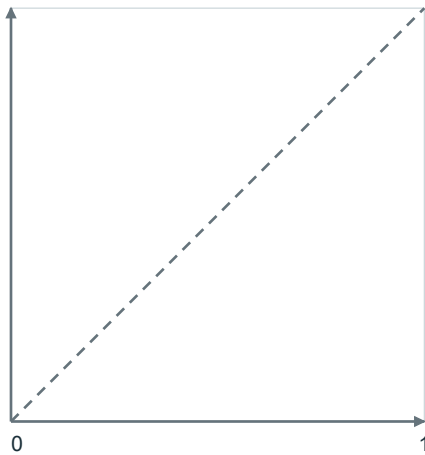
Will feedback find the point that exists?

Core: fractions and graphing

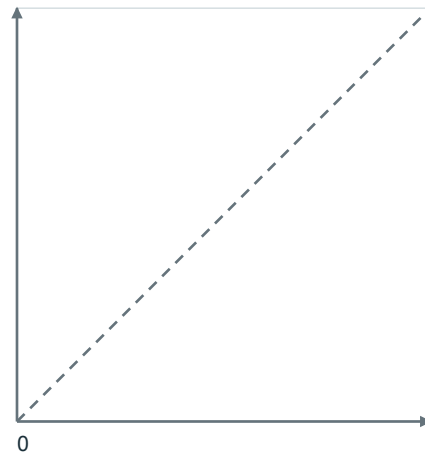
A machine takes x in $[0,1]$ and returns $f(x)$, also in $[0,1]$. Feedback repeatedly uses the output as the next input: $x_{(n+1)}=f(x_n)$. A fixed point is an input that comes out unchanged.

1. Choose a starting value and compare the machines $f(x)=(x+1)/3$ and $g(x)=1-x$. Record four updates. Find each machine's fixed points, then choose a start that makes their feedback behavior disagree.

$$f(x)=(x+1)/3$$



$$g(x)=1-x$$



Starting value / four updates / fixed points

2. For f , derive an exact formula for the error after n updates from any start x_0 . How many updates guarantee error below $1/100$ for every start in $[0,1]$? Does g have a similar decreasing bound?

Your construction / explanation

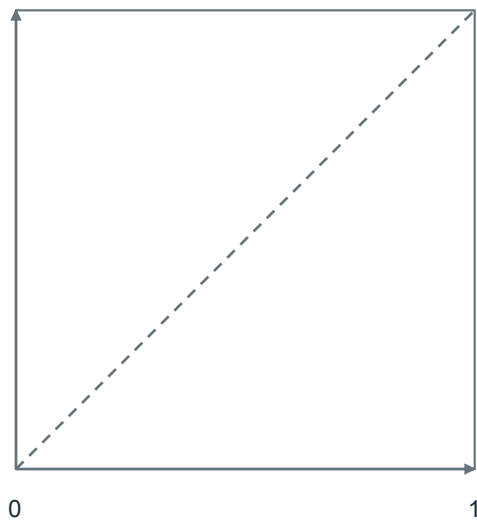
Prove existence without promising success

Will feedback find the point that exists?

Core continuation: a supplied continuity theorem

Supplied theorem: if a continuous real function has values of opposite signs at two endpoints, it has a zero between them. A zero at an endpoint already counts. Now f is any continuous function from the closed interval $[0,1]$ into itself.

3. Try to draw a continuous graph inside the unit square that never meets the diagonal $y=x$. Then prove why no such graph exists, including the endpoint cases.



Attempt / endpoint comparison / proof

4. Give one continuous interval machine with many fixed points and one with a unique fixed point whose feedback need not converge. Which claims did the existence proof leave open?

Your construction / explanation

A whole family of machines

Will feedback find the point that exists?

Continuation: powers, signs and limits

For $-1 \leq q \leq 1$, define $f_q(x) = 1/2 + q(x - 1/2)$. All these machines map $[0, 1]$ into itself. The parameter q can be chosen by the learner.

5. Classify the fixed points and feedback behavior for every allowed q . Explain exactly when errors alternate, shrink, stay unchanged or cycle. Include $q = -1, 0, 1$.



Choose parameters; find the cases that behave differently.

Fixed points / exact error formula / complete classification

6. Change the domain to the open interval $(0, 1)$ and use $h(x) = x/2$. It halves every distance. Does it have a fixed point in this domain? Explain why this does not contradict your closed-interval proof.

Your construction / explanation